# What Should Economics Ask Next?

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**Abstract**

Choosing what to work on is one of the least formalized decisions in science. I study whether the local structure of past research helps screen which still-open questions are worth reading first. I build a directed literature graph from 242,595 published economics-facing papers from 1976 to 2026, rank open questions using only the literature available at each date, and test those rankings against which questions later enter published work. On realistic shortlists, graph-based screening outperforms ranking by popularity alone, and a second-stage model built on the same graph improves further. The gains differ across two broad kinds of question. Questions that deepen existing claims with clearer mechanisms produce more later realizations per inspected shortlist, while questions that close missing direct relations cover a larger share of the links the literature eventually adds. More broadly, the historical record suggests that economics more often adds mechanisms around existing claims than closes locally implied direct relations. The paper therefore contributes in two ways. Practically, it shows that local literature structure contains useful screening information. Descriptively, it uses that screen to characterize how economics more often moves.

**Keywords:** research direction, economics of science, scientific discovery, question choice, literature graphs

## 1 Introduction

Choosing what to work on is one of the least formalized decisions in economics. We have disciplined tools for identification, estimation, and inference. We have much less guidance for the upstream choice of which question deserves scarce attention in the first place. Bloom et al. (2020) argue that ideas are getting harder to find, while Jones (2009) emphasizes the growing knowledge burden faced by new researchers. Those arguments point in the same direction: as the stock of published work grows, it becomes harder to see which question is worth pursuing next. Foster, Rzhetsky, and Evans (2015) add a second concern. Scientists overwhelmingly choose conservative strategies even though riskier exploration disproportionately generates high-impact work. The problem is not only that question choice is hard. It is also tilted toward the familiar.

That problem becomes sharper when AI lowers the cost of downstream research tasks such as writing, background research, data analysis, and coding (Korinek, 2023). Agrawal, McHale, and Oettl (2024) make the complementary point in a model of prioritized search: once ranking hypotheses becomes cheaper and better, the value of choosing what to test shifts further upstream. The question here is narrower than "what should economics study?" in the abstract. It is whether the structure of past research can help rank candidate next questions in a disciplined way.

This paper asks whether the structure of past research can help economists decide which open questions are worth reading first. I build a directed literature graph from published economics-facing papers and use it to surface two kinds of candidate next questions. In a direct-to-path case, the literature already contains a direct relation, but later work adds a clearer mechanism around it. In a path-to-direct case, the literature already contains nearby support for a relation, but the direct link itself is still missing and only appears later. The distinction matters because these are different scientific moves. The first asks where an accepted claim is still underexplained. The second asks where nearby evidence already points toward a direct relation that the literature has not yet stated explicitly.

An example makes the distinction concrete. Suppose the literature already links broadband access to business formation. Later work may connect that relation more clearly to search frictions, market access, or information costs. That is a direct-to-path case. Now consider a case where the literature already links trade liberalization to imported inputs and imported inputs to firm productivity, but has not yet stated a direct trade-liberalization-to-productivity relation. That is a path-to-direct case. The paper studies both prospectively in the same graph, while keeping them separate enough to compare.

The empirical comparison is simple. At each date, I freeze the literature at $t - 1$, rank still-open candidate questions using only the graph observed by that date, and then ask which of them later become part of published work. I compare a popularity baseline, a transparent graph score, and a learned reranker built on the same candidate set. The popularity baseline matters because it captures one plausible model of how scientific attention moves. If later work mostly follows visibility and existing traffic, popularity should already do well.

The main result keeps the two historical families side by side. The graph helps in both. The transparent score improves on the popularity baseline in the top 100 suggestions, and the learned reranker improves further. But the gains differ in form. Direct-to-path yields more later realizations per 100 suggestions, while path-to-direct captures a larger share of all later-realized links and ranks them earlier on average. The graph is therefore most useful as a tool for directing scarce reading time, not as a substitute for broad reading or field knowledge.

The paper answers two questions. First, does graph-based screening line up better with what later becomes part of the literature? Second, what does that screening exercise say about how economics tends to move? The core analysis starts with the paired shortlist comparison across the two historical families and then asks how often the literature deepens mechanisms around existing claims versus closing a missing direct relation. Appendix A reports the supporting material: extraction details, expanded model com-

parisons, paired extensions, credibility checks, supplementary usefulness exercises, and graph-evolution descriptives.

## 2 Related Literature

This paper speaks to four literatures.

First, it speaks to the economics of ideas, discovery, and research direction. Bloom et al. (2020) document rising research effort alongside falling research productivity across several domains, while Jones (2009) studies how accumulating knowledge changes the organization of innovative activity.[1] Those papers focus on the production of ideas and the cost of reaching the frontier. A related literature asks whether incentives and norms steer researchers toward safe topics. Azoulay, Graff Zivin, and Manso (2011) show that tolerating early failure produces more novel and more high-impact work. Foster, Rzhetsky, and Evans (2015) show that conservative strategies dominate in practice even though riskier exploration disproportionately generates breakthroughs.[2] The present paper studies a narrower but operationally central problem: given a large existing literature, can the structure of past research help screen candidate next questions in a way that is less path-dependent than pure cumulative advantage?

Second, the paper draws on science-of-science work that uses large-scale scientific data to study novelty, impact, and frontier formation. Fortunato et al. (2018) provide a broad synthesis. Wang and Barabasi (2021) provide a quantitative overview of that literature, while Uzzi et al. (2013) show that novelty often combines conventional structure with a limited number of atypical combinations. Park, Leahey, and Funk (2023) document a broad decline in disruptive science since the mid-twentieth century, and Wang, Veugelers, and Stephan (2017) show that genuinely novel papers face a short-run citation penalty even though they are far more likely to become top-cited in the long run.[3] That literature is relevant in spirit, but the empirical object here is different. I do not measure novelty from citations or reference-pair combinations. I define candidate next questions as missing links in a directed claim graph and evaluate them against later publication.

Third, the paper relates to network growth, link prediction, and learning to rank. Price (1976) and Barabasi and Albert (1999) show why already connected nodes tend to attract more links. In this paper that is the main null, not a decorative comparison. If the future of the literature is mostly a popularity process, then a rich-get-richer rule should perform well when the target is future edge appearance. Preferential attachment is therefore a serious benchmark because it encodes one plausible model of how scientific attention moves. The broader link-prediction literature (Liben-Nowell and Kleinberg, 2007; Martinez et al., 2016) and

[1] The "ideas are getting harder to find" framing is itself debated. Fort et al. (2025) argue that what is declining is the translation of ideas into measured growth, not the rate of idea generation itself.

[2] Packalen and Bhattacharya (2020) argue that citation-based incentives have further shifted scientists away from new ideas toward established topics, creating a self-reinforcing momentum that a screening tool could partially offset.

[3] The disruption-decline finding is itself debated. Petersen, Arroyave, and Pammolli (2024) argue that part of the measured decline reflects citation inflation rather than a real shift in innovativeness. For this paper, the relevant implication is narrower: whether or not measured disruption is declining, the literature's structure visibly thickens around existing claims over time, and that thickening is exactly what the path-development results in Section 5.2 measure.

the learning-to-rank tradition (Liu, 2009) show that learned models can improve on fixed heuristics. More directly, recent work treats missing links in scientific knowledge graphs as candidate research directions. Krenn and Zeilinger (2020), Gu and Krenn (2025), Sourati et al. (2023), and Rzhetsky et al. (2015) are the closest examples. The present paper differs in three ways: it uses a directed economics claim graph, it evaluates candidate questions prospectively in walk-forward vintages, and it interprets the ranking problem as one of scarce reading time rather than generic link prediction.

A nearby but distinct literature asks a different question: not which missing relation later appears, but which disconnected literatures or rising clusters are worth joining or monitoring. Swanson's classic literature-based discovery program asks whether separately developed literatures imply useful hypotheses that no single literature states explicitly (Swanson, 1986). Dynamic topic models, burst-detection systems, and emerging-topic maps instead track which themes are gaining salience (Blei and Lafferty, 2006; Chen, 2006; Small et al., 2014). Those systems are adjacent to this paper, but the object here is narrower and more economist-facing. I do not rank topics by growth or ask whether any bridge between disconnected literatures looks plausible in principle. I rank candidate directed research questions and evaluate them against later publication.

Fourth, the paper enters a fast-moving discussion around AI-assisted scientific work. Recent systems already help with tasks such as hypothesis generation, literature synthesis, and manuscript or workflow support (Zhang et al., 2025; Shao et al., 2025; Korinek, 2023).[4] Si, Yang, and Hashimoto (2024) find in a large-scale evaluation that LLM-generated research ideas are rated more novel, though slightly less feasible, than expert-generated ones. Agrawal, McHale, and Oettl (2024) formalize AI-assisted discovery as prioritized search over a combinatorial hypothesis space. d'Aquin (2025) argues more broadly that knowledge graphs are becoming scaffolds for AI-based scientific discovery.[5] My contribution is not a new general-purpose AI assistant and not a new claim-extraction model. The paper uses AI-extracted paper-level structure as an enabling layer, then asks an economics question: can that structure be turned into an inspectable, prospectively testable object for deciding what to read and work on next?

That positioning also helps distinguish this project from generic link prediction. Generic link-prediction work asks whether a missing edge will appear in a network. The closest social-science comparisons are Tong et al. (2024), who extract causal relations from 43,312 psychology papers and use knowledge-graph link prediction to generate hypotheses, and Lee et al. (2025), who construct a 10,490-triplet benchmark from 2,595 economics and finance studies to evaluate LLM causal reasoning. The present paper differs in scale, in evaluation design (prospective walk-forward backtesting rather than expert judgment or causal-reasoning benchmarking), and in the separation between the benchmarkable historical object and the

[4] Contemporary deployed examples include Refine (https://www.refine.ink/), the ICLR 2026 Reviewer Guide (https://iclr.cc/Conferences/2026/ReviewerGuide), Stanford Agentic Reviewer (https://paperreview.ai/tech-overview), and Project APE (https://ape.socialcatalystlab.org/). These are useful examples of the current tool landscape, but they are not treated as archival scholarly references in the bibliography.

[5] A related concern is that data-driven tools may narrow exploration rather than broaden it. Hoelzemann et al. (2024) show that when data highlights attractive but suboptimal paths, it can suppress breakthrough discovery. That is a "streetlight effect." A screening tool has to be honest about that risk. The attention-frontier and heterogeneity results in the appendix speak to it directly.

surfaced question. The paper is therefore best understood as economics-first metascience with a graph-based empirical object, rather than as a pure machine-learning exercise.

Taken together, these literatures frame the paper's central question. If the frontier is harder to navigate, if incentives bias researchers toward the familiar, and if AI is lowering the cost of downstream research tasks, then the upstream choice of what to work on becomes more important and more amenable to structured screening. The paper tests whether a graph-based screen can add value beyond cumulative advantage in that specific decision problem. Table 3 summarizes how this paper differs from the closest comparable systems.

## 3 Data, Extraction, and Normalization

This section defines the paper's main data object before the benchmark begins. The paper starts from a published-journal corpus, extracts a paper-level claim graph from each title and abstract, matches repeated labels across papers, and then ranks missing directed links for prospective evaluation. The selected journal corpus contains 242,595 papers from the top 150 core economics journals and the top 150 adjacent journals under the field-weighted citation impact selection rule. The sample spans 1976 to early 2026. Of those papers, 230,929 contain at least one extracted edge, yielding 1,443,407 raw extracted edges. After normalization and graph construction, the benchmark graph retains 230,479 papers and 1,271,014 normalized links. The frozen ontology is a separate object used for node matching and audit; it contains 154,359 rows. The next figures walk through the pipeline in order: one paper-level extraction, cross-paper matching into a shared graph, one live graph neighborhood, and the difference between the benchmark event and the richer question a reader inspects.

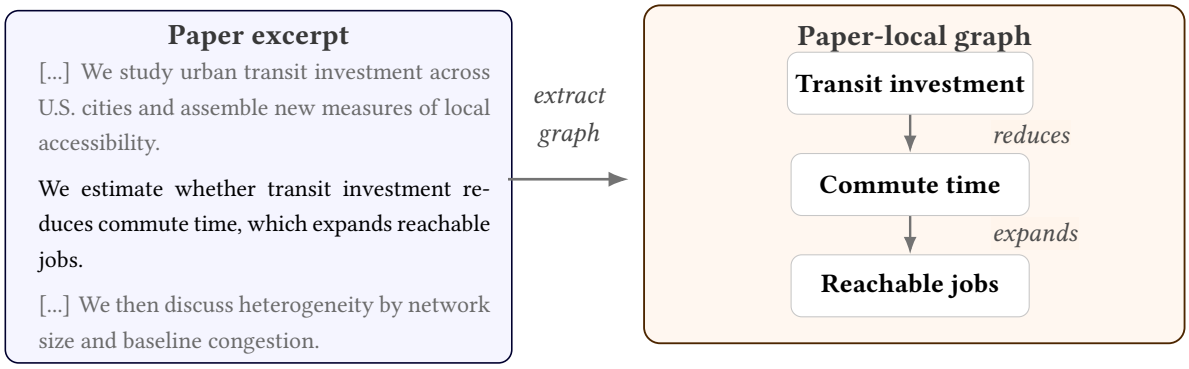
### 3.1 Corpus

The paper uses the published-journal corpus because the goal is to study realized scientific structure in an economics-facing literature, not to optimize coverage of working papers, drafts, or preprints. This choice is conservative. It sacrifices some freshness in exchange for clearer source control, more stable metadata, and a graph that is easier to interpret as realized economics research rather than a noisy mix of partially filtered text. Work-level metadata, source metadata, and journal assignments come from OpenAlex.[6]

The journal universe is deliberately narrow in two senses. First, it uses a selected journal universe rather than a universal bibliographic crawl. Second, it uses the FWCI-based selection rule so that the retained sample is tied to a published-literature core with credible economics or adjacent relevance. This is not meant to claim that the retained corpus exhausts the frontier. It is meant to define a reproducible, interpretable literature against which missing links can be backtested. Appendix G summarizes the resulting corpus waterfall and the retention numbers.

[6] OpenAlex enters the paper as the bibliographic and journal-metadata layer. The extraction, normalization, and ranking steps are built on top of that source rather than inherited from it.

**Figure 1:** A paper excerpt and its local graph




*Notes.* This figure isolates the extraction unit. One paper becomes one paper-local graph. The point is to show what is recovered before any cross-paper matching or normalization. The example is lightly cleaned from a live economics-facing transport-labor question, but the layout is stylized to make the extraction step legible.

### 3.2 Paper-local graphs

The extraction layer builds on Garg and Fetzer (2025). Each title and abstract is converted into a paper-level graph in which nodes are extracted concepts and edges summarize the relations the paper itself states, studies, or reports (Figure 1). The present paper extends that idea in three ways. First, the schema is broader than explicit causal claims, because the benchmark also needs undirected contextual support.[7] Second, the schema separates the paper's *causal presentation* from the *evidence method* used to support a claim. Third, the graph stores contextual qualifiers in dedicated fields rather than forcing them into the node label. The goal is not just to recover whether a paper makes a claim. It is to recover enough local structure that the claim can later be used inside a reusable concept graph. Code, prompt files, and release materials are available in the public repository at `https://github.com/prashgarg/frontiergraph`.

The decision to work paper by paper first is deliberate. At extraction time, the task is not to solve global concept matching inside the language model. It is to recover each paper's own internal concept reuse faithfully and to prevent the model from inventing relations that are only implied by transitivity. Exact prompts, the full schema, and the design logic are reported in Appendix D. **The code and prompt files used in this paper will be released at `https://github.com/prashgarg/frontiergraph`.**

### 3.3 Node normalization

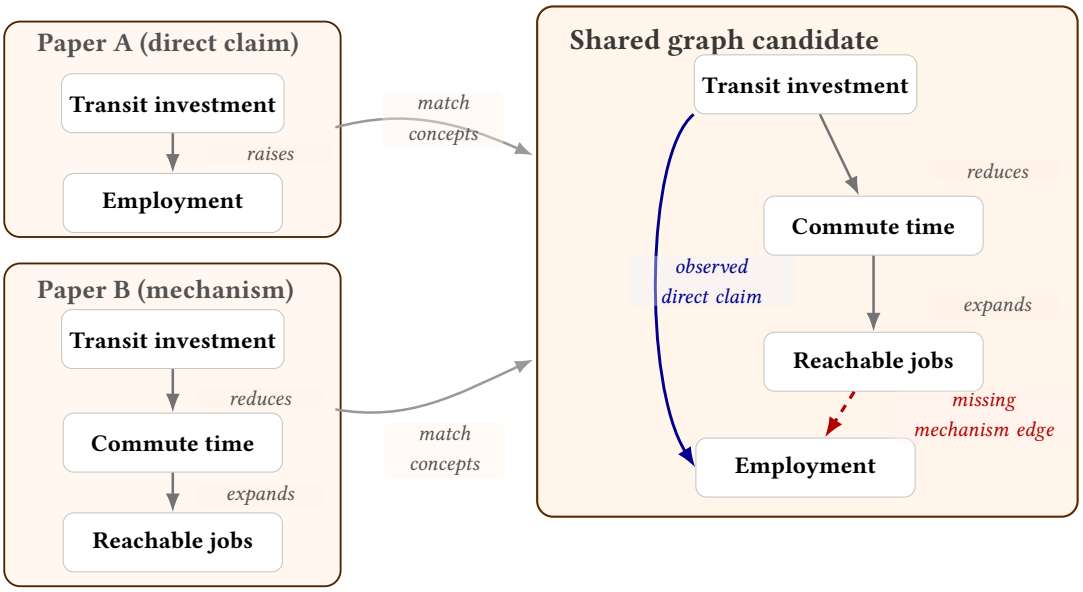
The normalization problem is central in this paper because candidate generation, path counts, gap measures, and missingness all depend on node identity. Paper-local concept strings vary in wording, scope, and granularity. "Inflation in Germany", "inflation", and "German inflation" should not automatically re-

[7] This is a deliberate departure from Garg and Fetzer (2025). Their object is the rise of credible causal language and design in economics, so the stricter identified-causal layer is the natural headline object there. Here the downstream task is broader research allocation over candidate questions, so the main target is a wider causal-claim layer, while the stricter identified-causal layer is retained as a nested credibility-oriented benchmark.

main three separate graph nodes if the downstream object is a reusable concept graph. But they should not be merged carelessly either.

For that reason the paper uses a fixed ontology and an auditable matching procedure rather than letting every paper-local label become its own global node or allowing unconstrained merging. The ontology combines JEL, an economics-filtered Wikidata pull, OpenAlex topics, OpenAlex keywords, and an economics-filtered Wikipedia crawl. The frozen baseline contains 154,359 rows. Matching then proceeds in stages. Exact lexical cases are resolved deterministically first. Harder cases use embedding retrieval, but the system keeps multiple candidates and reviewed overlays rather than forcing one irreversible merge.[8] The practical aim is simple: merge obvious duplicates, avoid false precision, and keep uncertain cases visible for later audit rather than hiding them inside one aggressive normalization pass. Appendix E gives the full algorithm and the retention counts. Table 1 fixes the key notation used throughout the paper.

**Figure 2:** Cross-paper matching reveals a shared candidate neighborhood




*Notes.* This figure isolates the cross-paper step. The two left panels are separate paper-local graphs. Only after repeated labels are matched into shared concepts does the right panel exist. The example is a direct-to-path case: one paper contributes a direct claim, another contributes a mechanism fragment, and the shared graph reveals a missing mechanism edge that can be benchmarked prospectively.

## 4 Question Objects and Evaluation

The graph is now a dated concept graph with directed causal links and undirected contextual support. Figure 3 shows one real neighborhood. The historical backtest needs a dated event. The reader needs a

[8] The embedding step is used as a retrieval and ranking device after exact lexical passes, not as an unconstrained merge rule. That ordering keeps obvious cases deterministic and makes softer matches auditable.

**Table 1:** Core notation used in the paper

| Symbol | Definition |
|---|---|
| $G_{t-1} = (V, E_{t-1})$ | Claim graph assembled from papers observed through year $t-1$. It contains directed causal links and undirected contextual support inside one concept-level graph object. |
| $u, v, w$ | Normalized concept nodes in the ontology-backed graph. |
| $u \to v$ | Directed causal link or directed causal candidate. |
| $\{u, v\}$ | Undirected noncausal pair. |
| $h$ | Evaluation horizon in years. |
| $K$ | Shortlist size in the fixed-budget retrieval problem. |

*Notes.* This table collects the symbols used repeatedly in the benchmark and scoring sections. The dated graph $G_{t-1}$ is always the graph observed through year $t-1$, so every score, feature, and candidate is constructed without seeing papers from year $t$ or later. $h$ is the forecast window in years, and $K$ is the number of candidate links or surfaced questions a reader is willing to inspect.

question worth inspecting. Those are not always the same object. I therefore separate two graph-grounded historical question objects from the richer reader-facing question built around them.

### 4.1 Two question objects

The graph supports two historical families of candidate question. In a *direct-to-path* case, the literature already contains a direct relation at date $t-1$, but the surrounding mechanism is still thin. Later work adds a mediating path around that relation. In a *path-to-direct* case, the literature already contains a local path at date $t-1$, but the direct relation itself is still missing. Later work states that direct relation explicitly.

The distinction matters because these are different scientific moves. Direct-to-path asks where an accepted relation is still underexplained. Path-to-direct asks where nearby evidence already points to a direct claim that the literature has not yet stated. Economists will often care more about the first object because much empirical work moves from reduced-form relations to channels, heterogeneity, and scope. But the second matters too because some later papers do close a missing direct relation rather than only elaborate an accepted claim.

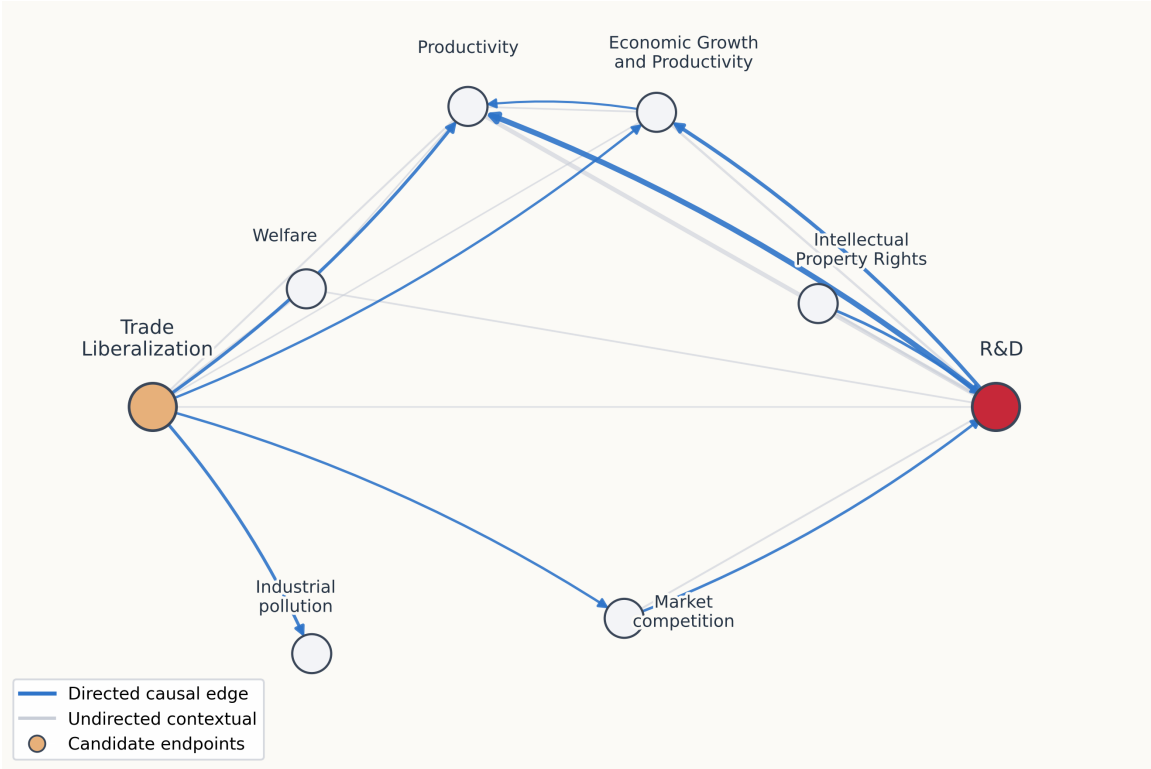
The headline benchmark is directed. At cutoff $t$, the candidate universe is the set of directed causal links that are still missing from the graph observed through year $t-1$. Undirected contextual support remains important because it helps score and interpret those directed candidates, but the main historical comparison does not rank undirected pairs separately.

For the directed causal task, the candidate universe at cutoff $t$ is

$$\mathscr{C}_t^D = \{(u, v) \in V \times V \,:\, u \neq v,\ (u \to v) \notin E_{t-1}^D\},$$

where $E_{t-1}^D$ denotes the directed causal subgraph observed through year $t-1$. A future realization occurs

**Figure 3:** A local neighborhood from the live graph




*Notes.* The previous figures are conceptual examples. This figure is a real neighborhood from the live graph, built around the candidate *Trade Liberalization* → *R&D*. Blue arrows are directed causal edges; gray lines are undirected contextual support. The figure is not a result in itself. Its job is to show that the benchmark ultimately operates on graph neighborhoods of this kind rather than on hand-drawn concept diagrams.

when $u \to v$ first appears during $[t, t+h]$. At each cutoff, take the active concepts, form all ordered source-target pairs, and drop any pair whose directed causal edge is already in the graph. What remains is the set of still-missing direct claims the paper can rank at date $t$. A candidate counts as realized only when that direct claim first appears within the horizon window, not when the concepts merely keep co-appearing or when the surrounding neighborhood changes in some other way.

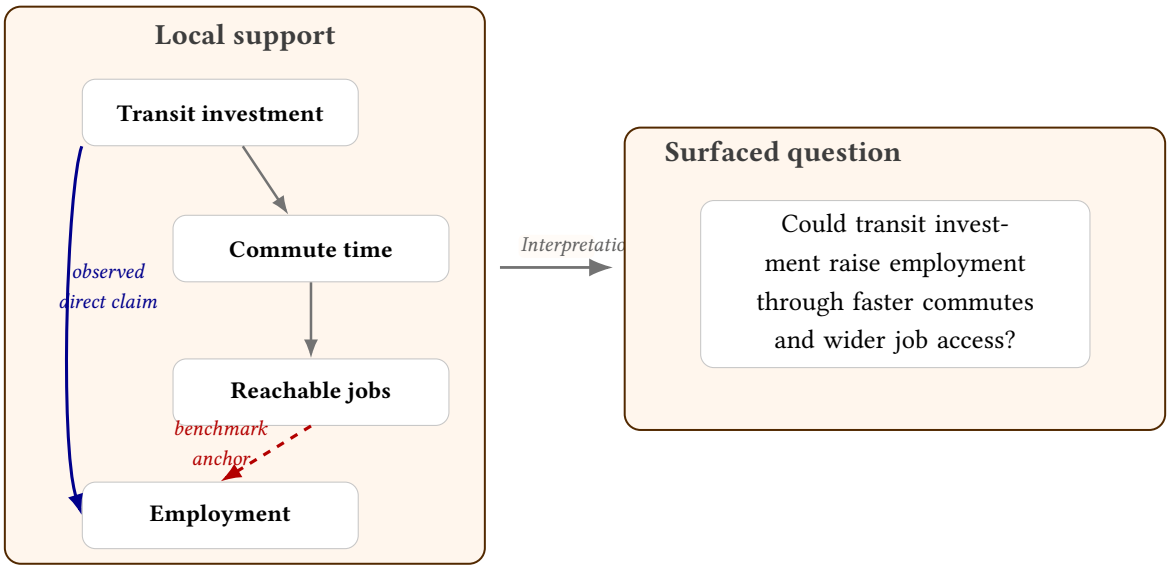
The historical event and the surfaced question are not the same. The historical event is whether the relevant direct edge or path appears later in the graph. The surfaced question is the human-readable prompt built around the local neighborhood. That separation matters because economists do not browse bare graph events. They browse questions that could plausibly become papers (Figure 4).

The surfaced question stays centered on the endpoints, or on the endpoints plus one mediator, for a practical reason. The benchmark needs one dated event, and the reader needs one inspectable question. A fuller local neighborhood may contain several paths and several overlapping readings, but those richer motifs enter as supporting evidence rather than as the event itself.

The main backtest also holds the active concept set fixed at the cutoff. A distinct extension is *node activa-*

*tion*: dormant ontology concepts, weakly grounded phrases, or genuinely new concepts entering the active graph later. That object matters, but it requires a different dated event than missing-link appearance and is therefore better treated as a separate frontier problem.

**Figure 4:** The benchmark anchor is narrower than the surfaced question




*Notes.* The dashed edge inside the left panel is the benchmark anchor that can be backtested. The reader-facing question is richer: it keeps that same anchor but presents the surrounding support graph as a readable question. The surfaced question is therefore a compression of the neighborhood evidence, not the benchmark event itself.

## 4.2 Gap and boundary questions

Two kinds of surfaced questions matter in practice. Gap questions already have rich nearby support but remain directly underworked. Boundary questions connect areas that still have little direct traffic between them. The paper's score is designed to separate these cases rather than collapse them into one novelty index. Gap questions are often easier to defend as plausible next questions; boundary questions are often more adventurous and potentially more fragile.

In the language of the graph, gap questions have high local support and low direct completion. Boundary questions have weaker local closure but connect parts of the graph that are still far apart. The distinction becomes substantively useful later, because the main model tends to surface a different gap-boundary mix than the popularity benchmark.

## 4.3 Transparent scoring rule

The ranking rule combines four ingredients: path support, underexploration gap, motif support, and hub penalty. Using the same transit-investment example that runs through the earlier method figures, path support asks whether the two endpoint concepts are already connected by short routes through nearby

mediators. The gap term asks whether those routes exist even though the direct relation itself is still absent or thin. Motif support asks whether the same endpoint pair keeps reappearing inside nearby structural patterns rather than in only one fragile corner of the graph. The hub penalty moves in the opposite direction: it reduces scores that are high only because both endpoints are extremely generic, heavily connected concepts.

Formally, the score for a candidate pair $(u, v)$ is a weighted sum of four components:

$$s(u, v) = \alpha \tilde{P}(u, v) + \beta G(u, v) + \gamma \tilde{M}(u, v) - \delta \tilde{H}(u, v),$$

where $\tilde{P}(u, v)$ is normalized path support, $G(u, v)$ is the underexploration gap, $\tilde{M}(u, v)$ is normalized motif support, and $\tilde{H}(u, v)$ is the normalized hub penalty. The main specification uses $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.3$, and $\delta = 0.2$. Those weights are fixed by design rather than tuned to maximize forecasting performance. A pair scores highly when several nearby routes keep pointing toward it, the direct relation still looks oddly absent relative to that support, and the pair is not simply another generic high-degree combination. Figures 5 and 6 are useful here because they show what the rule is actually rewarding: multiple short paths, repeated local support, and a direct link that still looks underfilled relative to its neighborhood. The notation is easier to read if each term is treated as answering one question. Is there already short local support connecting the endpoints? That is $\tilde{P}$. Does the direct claim still look missing relative to that support? That is $G$. Is the signal repeated in more than one nearby motif rather than hanging on one fragile path? That is $\tilde{M}$. Are the endpoints simply very generic hubs that would connect to almost anything? That is $\tilde{H}$, which is subtracted rather than added. So the score is not an opaque index. It is a transparent compromise between support, missingness, repeated local confirmation, and a penalty for generic popularity.

That transparency choice matters for interpretation. The graph already stores stability, causal-presentation, evidence-type, and edge-role metadata, but the main score does not yet fully weight those signals. That is deliberate in this version. The score is meant to be inspectable question by question rather than optimized as a forecasting black box, and richer credibility weighting is better understood as the next extension than as a hidden tuning layer.

### 4.4 Learned reranker

The transparent score is fixed-weight and inspectable. Its weights are chosen by design rather than tuned to historical outcomes, which keeps the score readable question by question. The cost is that fixed weights may miss combinations of graph features that matter for screening. The learned reranker asks a narrower question: if the candidate set and time discipline stay fixed, does a model that learns how to weight the same graph information improve the shortlist? The reranker uses the same missing-link candidates, only information available at the historical cutoff, and no text, author, or institutional features.

**Feature families.** The reranker uses five nested feature families. The first is the transparent graph score itself. The second adds local structure: path support, motif counts, mediator counts, co-occurrence signals, endpoint degree products, and a same-field indicator. The third adds timing: support age, recency of the most recent supporting edge, and recent-window degree and incident counts for each endpoint. The fourth adds evidence composition: mean edge stability, evidence-type diversity, venue diversity, source diversity, and mean field-weighted citation impact at each endpoint. The fifth adds boundary and gap indicators: whether the endpoints sit in different field groups with no co-occurrence, whether the pair looks gap-like, and the local closure density around the pair. The richest specification uses 34 graph-derived features. Table 14 in Appendix H lists the full inventory.

**Training design.** Training follows the same walk-forward discipline as the main benchmark. At each evaluation cutoff $t$, the model is trained only on earlier cutoff-year cells and is evaluated at $t$. Features are computed from the historical corpus through year $t-1$. The positive label is whether the candidate edge first appears within the evaluation horizon. I test two model families: logistic regression with class-balanced weights, and a pairwise ranking model trained on positive-versus-negative feature differences. Both use $L_2$ regularization. Because the feature families are nested, the paper can ask a simple question at each step: does adding more graph information improve screening beyond topology alone? Appendix H gives the full training design, regularization grid, best configurations by horizon, and grouped feature decompositions.

**Scope.** The reranker stays inside the graph-screening framing: every feature comes from the literature graph (no text, no author identity, no external data), the candidate set is the same set of missing links, and the temporal discipline is the same walk-forward design.[9]

### 4.5 Walk-forward evaluation

The prospective design freezes the graph at year $t-1$, ranks candidates using only information available at that date, and then checks whether those links first appear over the evaluation horizon. This vintage design avoids using future edges, future degrees, and later realizations when scoring the historical cutoff year,[10] which keeps the interpretation close to the real ex ante decision problem. The benchmark is therefore not "how well does the score fit the historical graph?" but "how well would this score have surfaced links that later became realized work?" That distinction matters because retrospective fit can reward mechanical regularities that are not actually useful for research allocation.

For each cutoff year, the graph is built from the historical stock only. Realizations are then defined from future papers over the chosen horizon. The benchmark is rolling rather than one-shot, so each horizon

[9] The difference is only in how the graph features are combined: by fixed design judgment in the transparent score, or by weights learned from earlier cutoffs in the reranker.

[10] In other words, the model at year $t-1$ is not allowed to borrow information from papers that appear in $t$ or later. That is the sense in which the design avoids leakage here.

is evaluated across multiple cutoff dates. A cutoff is eligible for horizon $h$ only if $t + h \leq 2026$, and the appendix horizon extension applies the same rule on a five-year cutoff grid when it extends the transparent benchmark to $h = 20$. The headline benchmark focuses on directed causal candidates, but the fuller atlas also evaluates directed and undirected objects separately and then pools them by weighted aggregation rather than by constructing one mixed ranking universe. Figure 7 makes the temporal discipline visual.

**Benchmark objects and metrics.** The paper keeps two dated historical objects side by side. In path-to-direct, a local path already exists and the later event is the first appearance of the direct link. In direct-to-path, a direct relation already exists and the later event is the first appearance of a supporting path. *Future links per 100 suggestions* is the shortlist hit rate: Precision@$K$, expressed per 100 suggestions so it stays comparable across reading budgets. Recall@100 asks what share of all later realizations in that family appear in the top 100. Mean reciprocal rank rewards putting later realizations nearer the top. These measures should be compared within family first, and then across families with the denominator difference in mind.

**Preferential attachment as benchmark.** The first benchmark is a cumulative-advantage rule. It asks whether one can do well simply by following existing traffic in the graph. A candidate looks better when it joins two endpoints that already attract many links. The standard network name for that logic is *preferential attachment*. In this paper it scores a candidate ordered pair by source out-degree times target in-degree:

$$\mathrm{PA}(u, v) = d^{out}(u) \times d^{in}(v).$$

It is a serious benchmark here, not a decorative one. Scientific attention is not allocated by neutral exploration alone. Visible topics, familiar data, established methods, and a larger installed literature often attract still more work. If a graph-based screen cannot beat that rule, then future question choice is mostly a popularity process. The appendix reports stronger transparent variants. The main text asks the simpler question of whether graph structure adds screening value beyond cumulative advantage.[11]

**Co-occurrence as a benchmark.** The second benchmark is a co-mention rule. It asks whether one really needs directed claim structure at all. Perhaps it is enough to know that two endpoint concepts appear together repeatedly in the same papers, even if one ignores direction, claim type, and local path structure. That is also a serious null in this setting. Many literatures are organized less by explicit causal direction than by repeated joint attention: topics, variables, and mechanisms that are often discussed together keep reappearing together. If that is the main force in the data, then a simple co-occurrence score should already recover much of what later gets studied. Much of the recent work on predicting

[11] The label "preferential attachment" comes from the network-growth literature. In that literature, nodes that already have many links are more likely to receive new ones (Price, 1976; Barabasi and Albert, 1999). Readers who do not use the term can think of it more simply as the graph analogue of cumulative advantage or a rich-get-richer process. In a science setting, the same logic says that already visible papers, concepts, or topics tend to attract still more connections. That is exactly why it is the right null in this paper.

future research connections takes exactly that route. Krenn and Zeilinger (2020) and Gu and Krenn (2025), for example, work with undirected co-occurrence, while related scientific-discovery systems often begin from co-mentioned entities or undirected knowledge links (Rzhetsky et al., 2015; Sourati et al., 2023). That approach has clear advantages: it requires no extraction of causal direction or claim metadata, and it scales easily. The question here is whether the harder extraction of directed claim structure adds screening value beyond what co-mention alone delivers. I therefore include a co-occurrence baseline that scores each candidate pair by how many papers mention both endpoints, ignoring direction, claim type, and path structure.[12]

Preferential attachment is the cumulative-advantage null. Co-occurrence asks whether undirected co-mention is enough. The transparent graph score is the readable graph baseline. The learned reranker is the strongest graph-based screen on the same candidate set.

**Fixed-budget retrieval.** The evaluation problem is not whether a useful question appears somewhere deep in a long ranking. It is whether useful questions appear in the first set a reader can plausibly inspect. Researchers do not read 5,000 candidates. Editors do not review 5,000 possibilities for a special issue. Funders do not investigate an unlimited menu of exploratory ideas. The relevant object is therefore a fixed reading budget: what does the graph deliver in the first 50, 100, or 500 suggestions that scarce attention can actually reach? That is why the paper emphasizes Recall@100, other fixed-budget shortlist measures, and frontier-style comparisons over larger $K$. Those metrics are not a cosmetic presentation choice. They correspond to the real screening problem the paper is trying to inform.

**Horizon choice.** The main horizons are 5, 10, and 15 years. Five years is a natural first window in economics because publication and diffusion are slow relative to many neighboring fields (Hadavand et al., 2024). But a five-year window is not enough to see many slower adjustments in topic, method, and citation uptake. Economics papers often keep accumulating attention over long citation life cycles, which is one reason to look beyond only the short run (Hamermesh, 2018). Ten years therefore captures slower movement in questions that take time to propagate through papers, methods, and field conventions. Fifteen years remains empirically useful because many slower literatures are still only partly visible at shorter windows. Three years and twenty years are kept as appendix extensions, but the main benchmark comparison is now centered on 5, 10, and 15 years.

[12] I do not know of a clean economics paper that uses this exact co-occurrence rule as a prospective benchmark for open-question screening, so I do not force an economics citation here. In economics and adjacent bibliometric work, co-mention, co-citation, and keyword co-occurrence are more often used descriptively to map fields than prospectively to rank next questions. The role of the benchmark here is narrower: it asks whether repeated joint mention alone is already enough, or whether directed claim structure adds screening value on top of that.

# 5 Results

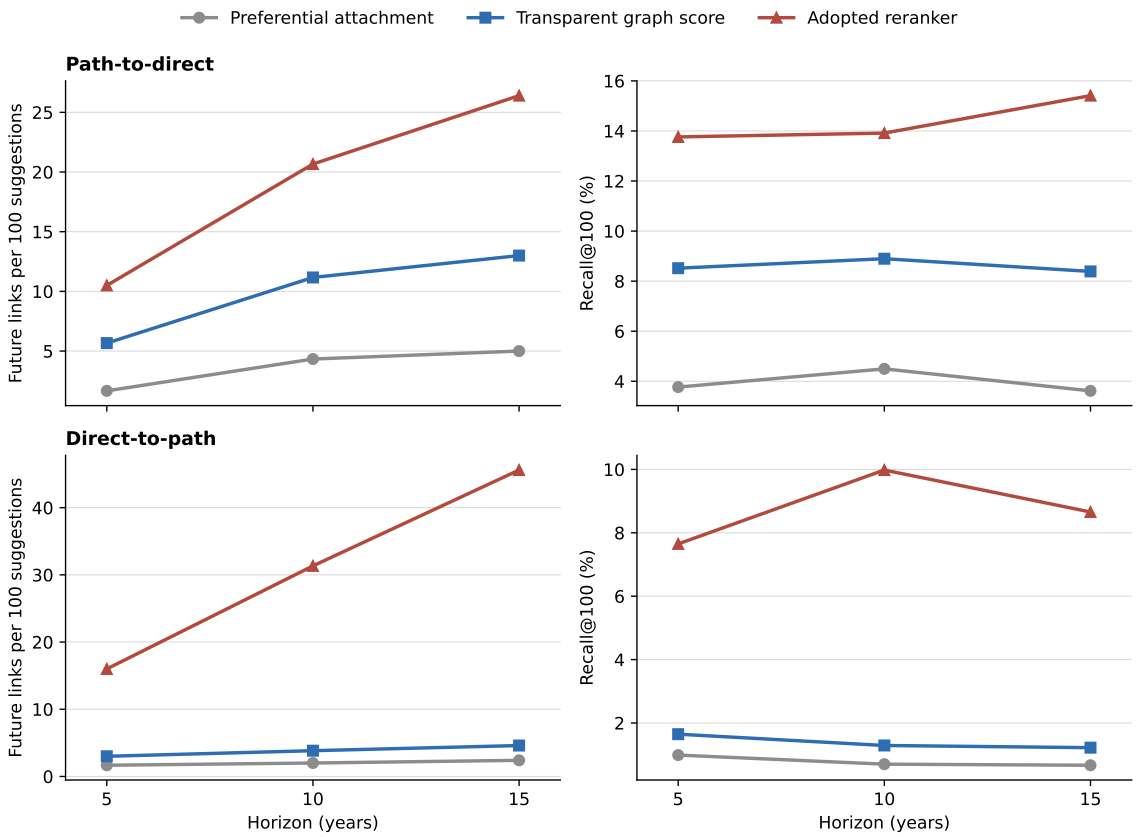
The results reported first are the ones available symmetrically for both historical families. The families are related but not identical. Path-to-direct asks whether a locally supported direct relation later becomes explicit. Direct-to-path asks whether an existing direct relation later acquires a clearer mechanism. The same shortlist metrics can be used for both, but they answer different questions. Within each family, the metrics describe how well the ranking surfaces later realizations of that family-specific event. Cross-family comparisons are still useful, but they should be read as comparisons across different historical motions rather than as one common denominator problem. Family-specific frontier, weighting, and heterogeneity extensions stay in the appendix until both families have matching runs.

## 5.1 Paired shortlist comparison

The first comparison asks whether local structure adds anything beyond popularity. That is a real test in this setting because research attention is path-dependent. Visible topics, familiar data, and established methods tend to attract still more work. If that cumulative-advantage process does most of the work, a popularity rule should already rank future questions well. It does not. In both historical families, the transparent graph score beats preferential attachment, and the adopted reranker improves further.

The two families then diverge in a useful way. Direct-to-path yields denser shortlists. In plain language, a larger share of the top 100 suggestions later becomes part of the published record. At $h = 5$, the adopted reranker surfaces about 16.0 future links per 100 suggestions in direct-to-path, versus 10.5 in path-to-direct, which is equivalent to a 16.0% versus 10.5% shortlist hit rate. At $h = 10$, the comparison is 31.3 versus 20.7. At $h = 15$, it is 45.6 versus 26.4. Path-to-direct instead captures more of the later-realized stock and ranks it earlier on average. The adopted reranker reaches Recall@100 of 13.8, 13.9, and 15.4 across $h = 5, 10, 15$, versus 7.6, 10.0, and 8.7 in direct-to-path (Figure 8 and Table 2).

**Figure 8:** The two historical families are informative in different ways




*Notes.* Each row is one historical family and each column is one shortlist metric. Path-to-direct asks whether a locally supported direct relation later appears. Direct-to-path asks whether an existing direct relation later acquires a supporting path. Within each family, the transparent score beats preferential attachment and the adopted reranker improves further. Across families, the comparison is not one-dimensional: direct-to-path delivers more later realizations per 100 suggestions, while path-to-direct has higher Recall@100 and MRR.

This is a severe screening problem in both families. The top-100 shortlist is being asked to pull forward a small fraction of a much larger future stock. The graph helps, but it helps in different ways. In direct-to-path, the main gain is a denser flow of later realizations through the shortlist. In path-to-direct, the main gain is bringing a larger share of the later-realized stock nearer the top of the ranking.

That distinction is useful rather than awkward. It says that the two families should not be collapsed into one generic score. They capture different forms of research progress. One is the later appearance of a direct relation. The other is the later appearance of a mechanism around an existing relation. Economists will often care more immediately about the second object, because much empirical work moves from a reduced-form relation to channels, heterogeneity, and scope conditions. But the first still matters, because some later papers do close a missing direct claim that nearby support had already made plausible.

## 5.2 Mechanism thickening versus direct closure

### 5.2.1 Aggregate patterns

This subsection asks which motion is more common. One is direct closure: a supporting path already exists at $t-1$, the direct link does not, and that direct link later appears. The other is mechanism thickening: a direct link already exists at $t-1$, but a supporting mediator path appears only later. The distinction matters because it separates two uses of research effort. Direct closure states a missing claim. Mechanism thickening explains, disciplines, or extends a claim the literature already treats as meaningful. Work on conservative search and declining disruption suggests that later work often develops accepted lines rather than replaces them with wholly new ones (Foster et al., 2015; Park et al., 2023). I compare the two motions on the same dated length-2 graph structure (Figure 4).

The aggregate comparison is clear (Figure 9). Direct-to-path transitions dominate path-to-direct transitions in every cutoff-period block and at every horizon. The cleanest way to state this is in transition rates, because the eligible sets differ. At $h = 10$, the direct-to-path transition share rises from about 0.009 in the 1990s to 0.043 in the 2000s and 0.121 in the 2010s, while the corresponding path-to-direct shares are about 0.003, 0.007, and 0.014.

The gap is large. Economics more often elaborates mechanisms around claims it already states than closes a missing direct relation implied by nearby support. That is consistent with a broader view of scientific change in which the literature more often thickens around existing structure than opens wholly new direct connections (Foster et al., 2015; Park et al., 2023). In this paper's terms, the graph more often surfaces structured elaborations of accepted claims than missing direct links between otherwise distant concepts. That helps explain why direct-to-path is the more natural reader-facing object in economics: it lines up with a literature that often asks not only whether a relation exists, but through what channel, under what conditions, and with what supporting mechanism.

The pattern varies across the literature in economically interpretable ways. Adjacent journals are relatively more path-closure heavy than the core, but direct-to-path still dominates in both tiers. At $h = 5$, the share of realized transitions taking the path-to-direct form is about 0.45 in adjacent journals versus 0.28 in the core. Finance remains more direct-to-path heavy throughout. Figure 10 reports the journal-tier split.

The aggregate transition pattern also clarifies how the surfaced object should be read. The most useful question is often richer than a single missing edge, because the literature more often deepens mechanisms around existing claims than it closes one missing direct relation. Illustrative surfaced examples therefore belong naturally in a section that keeps both families in view rather than treating direct closure as the default object.

Taken together, the paired evidence says two things. In both historical families, graph structure improves on popularity and the reranker improves further. But the larger substantive point is that much of scientific development takes the form of mechanism-deepening around existing claims. The most useful next question is therefore often richer than a single missing edge.

### 5.3 Reranked results in both families

The baseline comparison is now established. The next question is whether learned graph features improve on the same candidate set once the transparent ranking has done the first pass. They do in both families. The reranker keeps the ontology, sample, and timing discipline fixed and changes only how graph features are weighted. The best reranker is not the same in every family and horizon, so the appendix reports the full tuning table. The main point is simpler: once the graph defines the candidate set, learned weighting improves on the transparent score in both families.

The learned reranker's role is the same in both families. It changes *which* questions move toward the top of the list. The transparent score remains useful because it explains *why* those questions look promising in graph terms: path support, missingness relative to nearby support, topology, and provenance. Table 2 summarizes the paired comparison.

**Table 2:** Paired benchmark summary for the two historical families

| | | Future links per 100 | | | Recall@100 | | | Mean reciprocal rank | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Family | Model | $h$=5 | $h$=10 | $h$=15 | $h$=5 | $h$=10 | $h$=15 | $h$=5 | $h$=10 | $h$=15 |
| Path-to-direct | Preferential attachment | 1.7 | 4.3 | 5.0 | 3.8 | 4.5 | 3.6 | 0.0021 | 0.0020 | 0.0019 |
| Path-to-direct | Transparent graph score | 5.7 | 11.2 | 13.0 | 8.5 | 8.9 | 8.4 | 0.0078 | 0.0078 | 0.0073 |
| Path-to-direct | Adopted reranker | 10.5 | 20.7 | 26.4 | 13.8 | 13.9 | 15.4 | 0.0133 | 0.0103 | 0.0116 |
| Direct-to-path | Preferential attachment | 1.7 | 2.0 | 2.4 | 1.0 | 0.7 | 0.7 | 0.0012 | 0.0010 | 0.0011 |
| Direct-to-path | Transparent graph score | 3.0 | 3.8 | 4.6 | 1.6 | 1.3 | 1.2 | 0.0019 | 0.0015 | 0.0015 |
| Direct-to-path | Adopted reranker | 16.0 | 31.3 | 45.6 | 7.6 | 10.0 | 8.7 | 0.0050 | 0.0087 | 0.0057 |

*Notes.* Each row reports one family-model combination averaged across the 1990–2015 cutoff grid. Columns are grouped by metric, with the three main horizons nested underneath. Future links per 100 suggestions is the shortlist hit rate, that is, Precision@100 expressed in percent. Recall@100 is the share of all later realizations in that family captured in the top 100. Mean reciprocal rank rewards putting later realizations nearer the top. The table shows the same pattern as Figure 8: the graph beats preferential attachment in both families, the adopted reranker improves further, and the two families remain substantively different rather than collapsing to one ranking problem.

The paired shortlist comparison already shows that graph structure adds screening value beyond popularity in both historical families. The reranker shows that the same candidate universe can be ordered better still. Additional paired extensions, including the current frontier and broader reading budgets, are therefore follow-on checks rather than the main claim (Appendix B and Appendix B.3). The paired budget comparison adds one practical lesson: path-to-direct is sharper when the reader wants a very short shortlist, while direct-to-path becomes more useful once the reader is willing to inspect a broader set of candidates (Appendix B.3). The separate path-length axis is useful, but it is parked off-paper until the main draft is settled.

## 6 Discussion and Extensions

The paper makes three claims. First, graph structure improves on a pure popularity rule in both historical families. Second, a richer weighting of the same graph features improves further. Third, the literature more often adds mechanisms around existing claims than it closes missing direct relations implied by nearby support. The point is not only that a graph can beat popularity. It is that local literature structure changes what economists should inspect first.

**What the distinction means.** The two historical objects capture different parts of research progress. Direct closure is the case where nearby support already points toward a relation that later becomes explicit. Mechanism thickening is the case where a known relation later acquires a clearer channel. Economists care about both, but they help in different ways. Direct-closure questions are useful when a researcher, editor, or funder wants a sharper shortlist of claims that look paper-ready but still underworked. Mechanism-thickening questions are useful when the task is to see where an accepted result is still structurally thin. The historical record says the second move is more common. A useful screening tool should therefore not only point toward underworked direct relations. It should also help identify where an accepted claim still lacks a convincing mechanism.

**Why to trust the pattern.** One reason to take that result seriously is that it does not rest on one fragile ranking rule. The main paired comparison survives the move from a transparent score to a learned reranker, and the reranker carries forward to later vintages that were not used for model selection (Figure 11). The appendices show the same discipline more broadly: the extraction layer is auditable, the directed claim graph is not just undirected co-mention in disguise, and the natural extensions confirm rather than reverse the main pattern.[13]

**Where the graph helps.** Two broader implications follow. First, the graph does not add the most value where the literature is maximally thin. The appendix regime splits show that its relative edge is larger where some usable local structure already exists. So this is not a machine for discovering the most remote idea in the graph. It is a screening device for the large middle of the literature, where there is enough nearby evidence to make a question legible but still enough missing structure that attention can be reallocated productively. Second, if the literature usually moves by thickening mechanisms around existing claims, then one useful role for computational tools is not open-ended idea generation. It is to help economists see where an accepted relation is still structurally underexplained.

**Benchmark event and surfaced question.** That is also why the paper keeps the benchmark and the reader-facing question separate. A dated graph event is useful because it can be tested prospectively. A

[13] The held-out temporal test is reported in Appendix H.8. The author-positioning null result is discussed in Appendix H. Credibility and stability audits are collected in Appendix I.

surfaced question is useful because it can be read, argued over, and discarded if it is uninteresting. The historical benchmark gives discipline. The surfaced object gives the tool practical value. Supplementary human and LLM usefulness checks remain secondary for that reason: they help assess whether the surfaced questions are readable, but they do not replace the historical evidence.[14]

**Budget interpretation.** The budget results sharpen that interpretation. Scarce attention is not only an individual researcher's problem. It is also an editorial, funding, and field-level problem. On that margin, the two historical families do not play the same role. Path-to-direct is more useful for triage: it concentrates more realized direct closures near the top of a short list. Direct-to-path is more useful for exploration: once the reading budget widens, it recovers a much broader share of the future mechanism literature. That is the relevant object when the problem is to see where an accepted relation is still underexplained rather than merely underclaimed.

That point also connects the paper to a broader literature on how science allocates attention. Jones (2009) and Bloom et al. (2020) stress the rising burden of reaching the frontier. Azoulay, Graff Zivin, and Manso (2011), Foster, Rzhetsky, and Evans (2015), and Packalen and Bhattacharya (2020) emphasize that institutions and norms can tilt scientific attention toward safer, more familiar directions. The present paper does not solve that problem at the level of incentives. But it does speak to scientific method in a narrower way: under severe screening constraints, a useful method is one that moves more plausible questions into the stage where economists, editors, and funders can exercise judgment. The graph is not a replacement for judgment. It is a disciplined way to widen the agenda that reaches judgment.

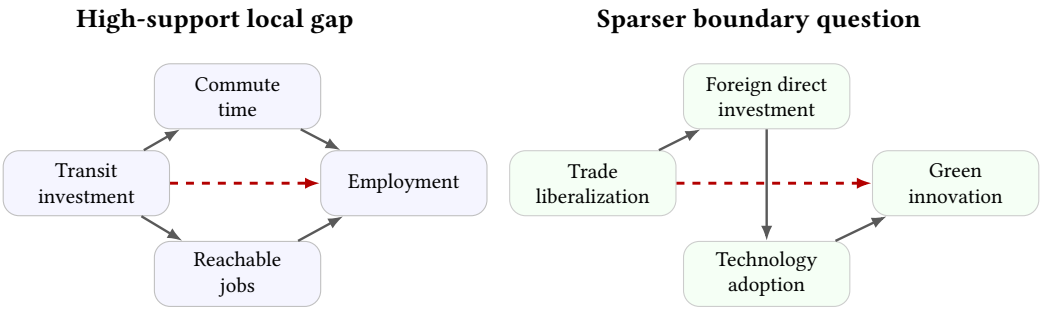
**Bundle uptake.** One natural extension is to ask whether later papers usually realize one predicted edge or a small bundle of connected predicted ideas. Work in science-of-science has emphasized that novel contributions often recombine nearby ingredients rather than arriving as isolated moves (Uzzi et al., 2013; Foster et al., 2015; Fortunato et al., 2018). The bundle-uptake analysis in Appendix K.1 suggests that the dominant downstream object in this setting is still a single paper-shaped question: only about five percent of realizing papers take up more than one historically predicted edge, and mixed-family bundles are very rare. But when multi-edge uptake does occur, it is strongly local in graph space. That result helps interpret the budget evidence. The wider-budget advantage of *direct-to-path* seems to come mainly from broader coverage across many separate papers, not from a large number of later papers absorbing broad mixed bundles at once.

The contribution here is modest but pointed: a prospectively testable historical design, a readable surfaced object, and an honest comparison with the cumulative-advantage null. Economics should not decide what to ask next only through popularity. The nearby structure of the literature carries screening information that popularity alone misses. If ideas are harder to find and downstream research tasks are getting cheaper, then better upstream screening becomes more valuable. The point is not to replace judgment. It is to widen

[14] Appendix J reports a small blinded human exercise and a larger appendix LLM screen. Both are useful as screening checks, not as substitutes for the historical benchmark.

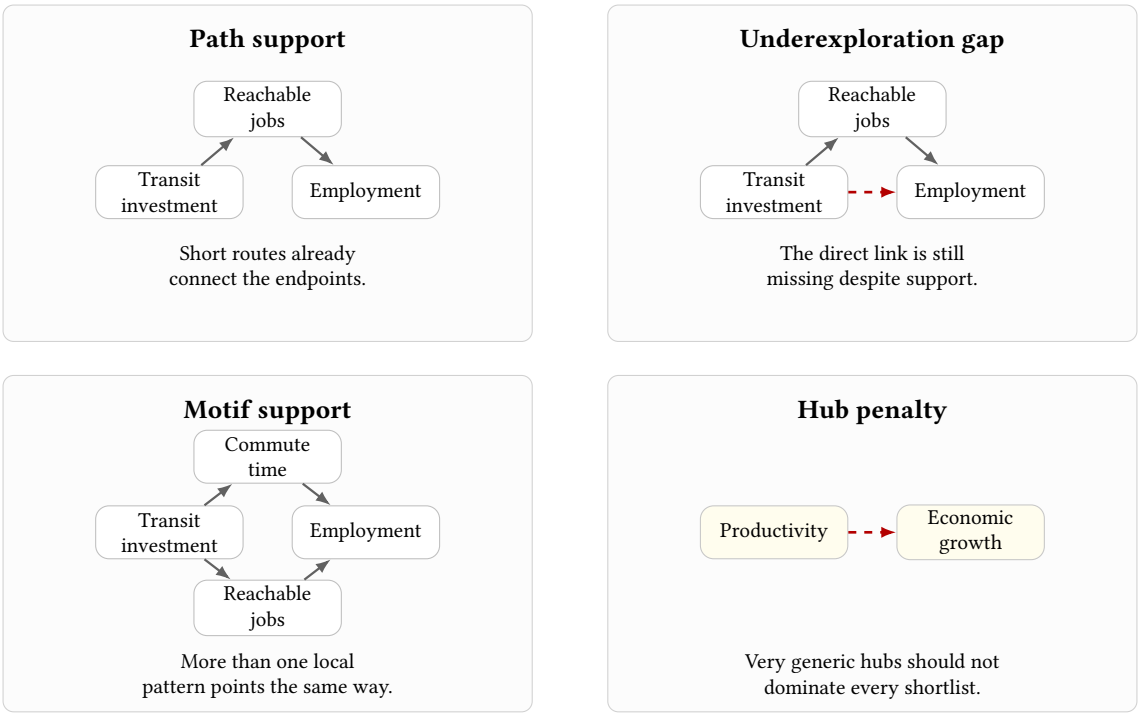
the set of questions that reaches the judgment stage.

**Figure 5:** Gap and boundary questions are different local graph patterns



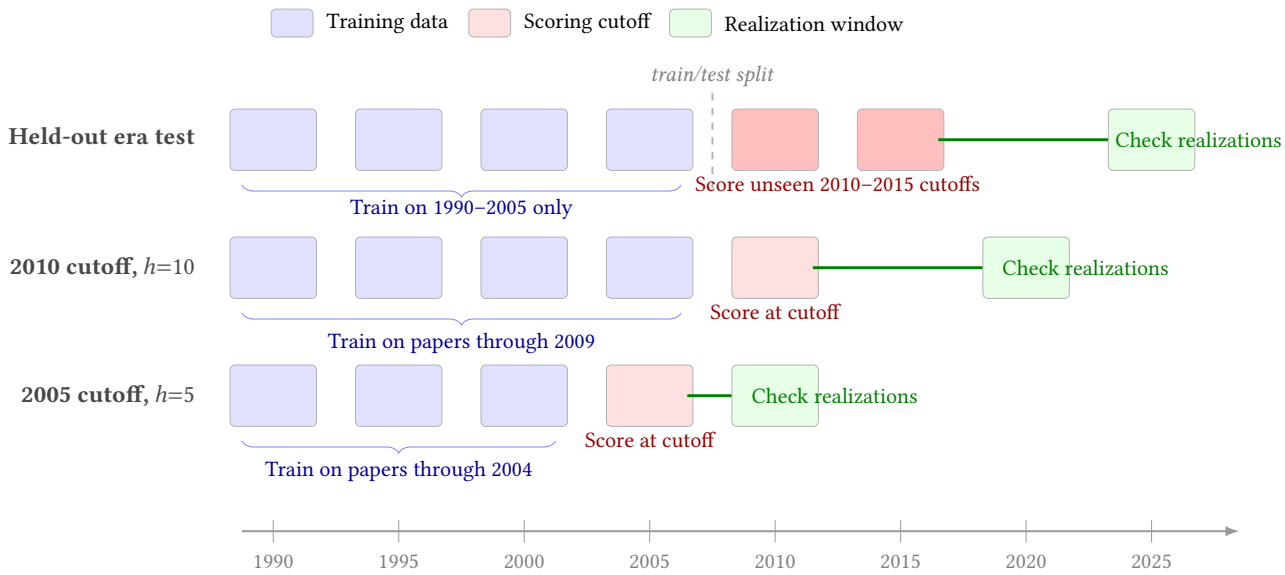

*Notes.* This figure exists to separate two screening cases that would otherwise look similar in prose. The left panel shows a gap-like candidate: nearby support is already dense, but the direct relation remains missing. The right panel shows a more boundary-like candidate: the two end concepts are connected only by a thinner mechanism chain. The node labels are hand-curated economics-facing examples used only to make the local graph pattern readable. The diagrams are conceptual rather than exhaustive.

**Figure 6:** The transparent score rewards supported but still undercompleted pairs



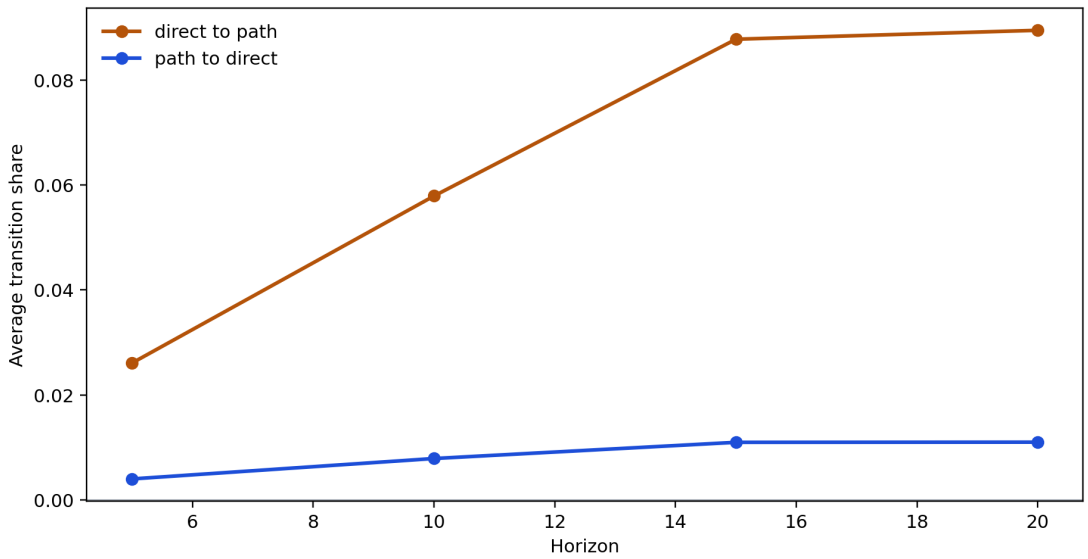

*Notes.* This figure makes the transparent score legible without the formula. The first three panels continue the same transit-investment worked example used in the earlier method figures. Path support asks whether short routes already connect the endpoints. The underexploration gap asks whether the direct relation is still missing despite that support. Motif support asks whether more than one nearby pattern points toward the same endpoint pair. The final panel intentionally switches to generic hubs to show the opposite force: the score should not rank a candidate highly only because both concepts are broad and heavily connected.

**Figure 7:** Walk-forward evaluation keeps scoring vintage-correct



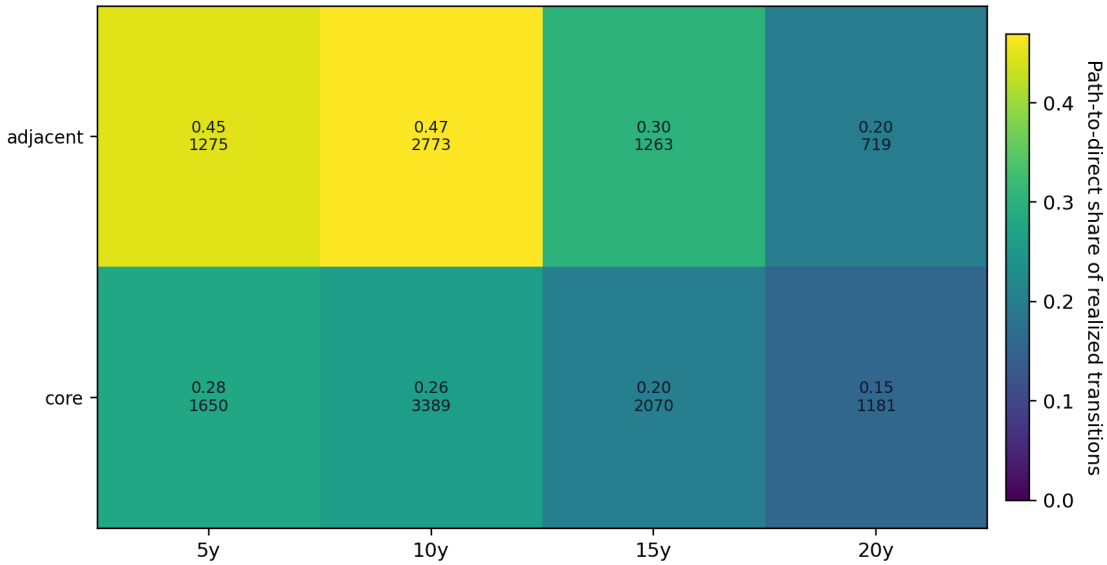

*Notes.* This figure exists to show the paper's temporal discipline visually. At each cutoff, the graph uses only papers published before that date (blue). Candidates are scored at the cutoff (red), and realizations are checked over the horizon window (green). The bottom row shows the held-out era test from Appendix H.8: train on 1990–2005 and evaluate on the unseen 2010–2015 era.

**Figure 9:** Mechanism thickening is more common than direct closure



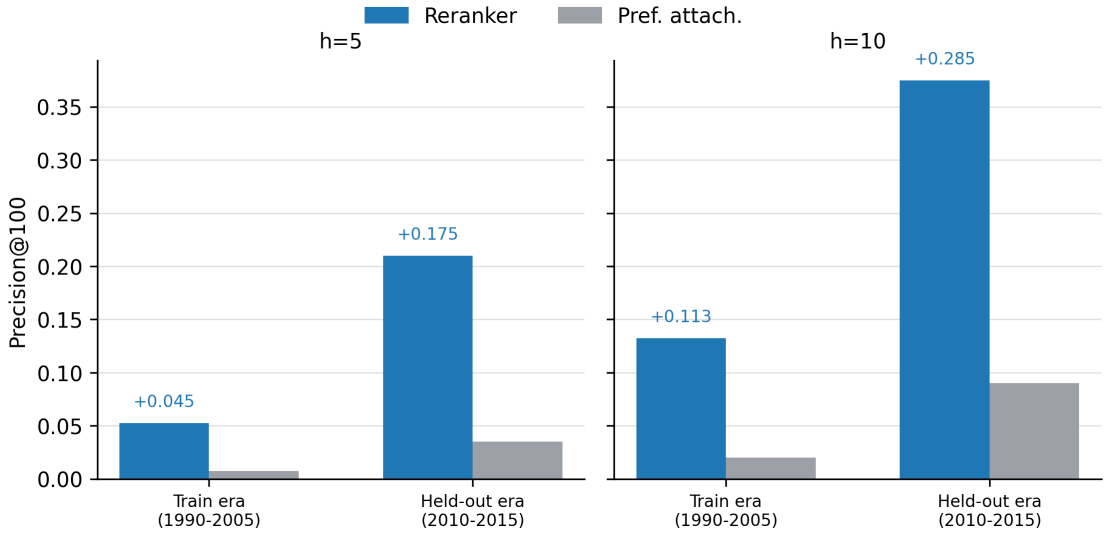

*Notes.* "Path to direct": a local path exists first and the direct link appears later. "Direct to path": the direct link exists first and a mediating path appears later. The figure exists to compare these two motions directly. Mechanism-deepening around existing claims dominates at all horizons.

**Figure 10:** Adjacent journals close more path-implied links, but direct-to-path still dominates




*Notes.* This figure asks whether the transition mix differs by journal tier. It reports the share of realized path-related transitions taking the path-to-direct form by journal tier and horizon. Adjacent journals are consistently more path-closure heavy than the core, but direct-to-path remains the larger transition type in both tiers.

**Figure 11:** The reranker generalizes forward in time




*Notes.* For each main horizon, the reranker is selected using only 1990–2005 cutoff cells and then evaluated on the fully held-out 2010–2015 era. Labels report the reranker's absolute $P@100$ gap relative to preferential attachment. The held-out gap is larger than the earlier training-era gap at both $h = 5$ and $h = 10$. Full results in Appendix Table 17.

# Appendix

What Should Economics Ask Next?

**Table 3:** Positioning relative to closest comparable work

| | This paper | Impact4Cast | Sourati et al. | Tong et al. |
|---|---|---|---|---|
| Domain | Economics | All sciences | Biomedicine | Psychology |
| Edge type | Directed causal | Co-occurrence | Co-occurrence | Causal |
| Corpus | 242,595 papers | 2.4M papers | Varies | 43,312 papers |
| Main null | Preferential attachment | ML baselines | Content-only | TransE |
| Temporal eval | Prospective walk-forward | Holdout | Holdout | None |
| Human eval | Appendix checks | No | No | Yes |

*Notes.* Impact4Cast: Gu and Krenn (2025). Sourati et al.: Sourati et al. (2023). Tong et al.: Tong et al. (2024). The table compares the closest systems along the dimensions most relevant for this paper: domain, edge type, scale, main null, temporal design, and whether any human check is reported.

# A Guide to the Appendix

The appendix has five jobs. First, it documents how the graph is built from title and abstract text. Second, it records the benchmark tables and reranker detail that support the main text without carrying its argument. Third, it collects paired and family-specific extensions that are useful once the main comparison is clear. Fourth, it reports audit and presentation checks on the graph and on the surfaced questions. Fifth, it adds a short set of supplementary analyses on how later papers realize predicted ideas and which kinds of teams move toward them. The main text remains narrow: paired shortlist results first, then the path-versus-mechanism result, then the confirmatory reranker comparison.

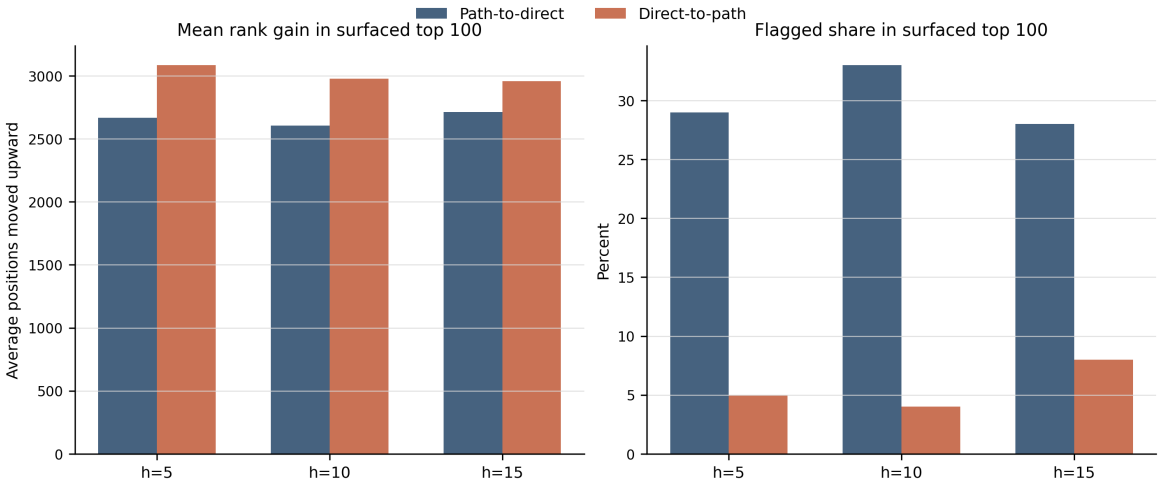
# B Paired family extensions

This section extends the paired comparison beyond the strict historical shortlist. The two figures below keep path-to-direct and direct-to-path side by side on two additional objects that are already available for both families: the current frontier and the broader-shortlist heterogeneity splits. Family-specific extensions that still exist only for direct closure remain in the next section.

## B.1 The two families surface different current frontiers

The historical benchmark and the current frontier answer different questions. The historical benchmark asks whether later papers validate a proposed relation or mechanism. The current frontier asks what each family would surface now after the reranker has reordered the large candidate pool. The paired current-frontier summary therefore compares the two families on the same simple diagnostics: how far the reranker moves surfaced items upward in the pool, and how often the surfaced top 100 still contains artifacts or generic endpoints.

**Figure 12:** The current frontier differs across the two question families




*Notes.* Each bar summarizes the surfaced top 100 for one family and one horizon on the current frontier pool. The left panel reports the mean upward movement in rank after reranking relative to the raw transparent ordering. Larger values mean the reranker is finding good candidates much deeper in the pool. The right panel reports the share of surfaced top-100 items that trigger the paper's artifact or generic-endpoint flags. The comparison is descriptive rather than historical. Its purpose is to show that the two families are not cosmetic variants of the same current shortlist. Path-to-direct keeps a larger flagged share in the surfaced top 100, while direct-to-path looks cleaner on that margin and still receives large upward rank gains from the reranker.

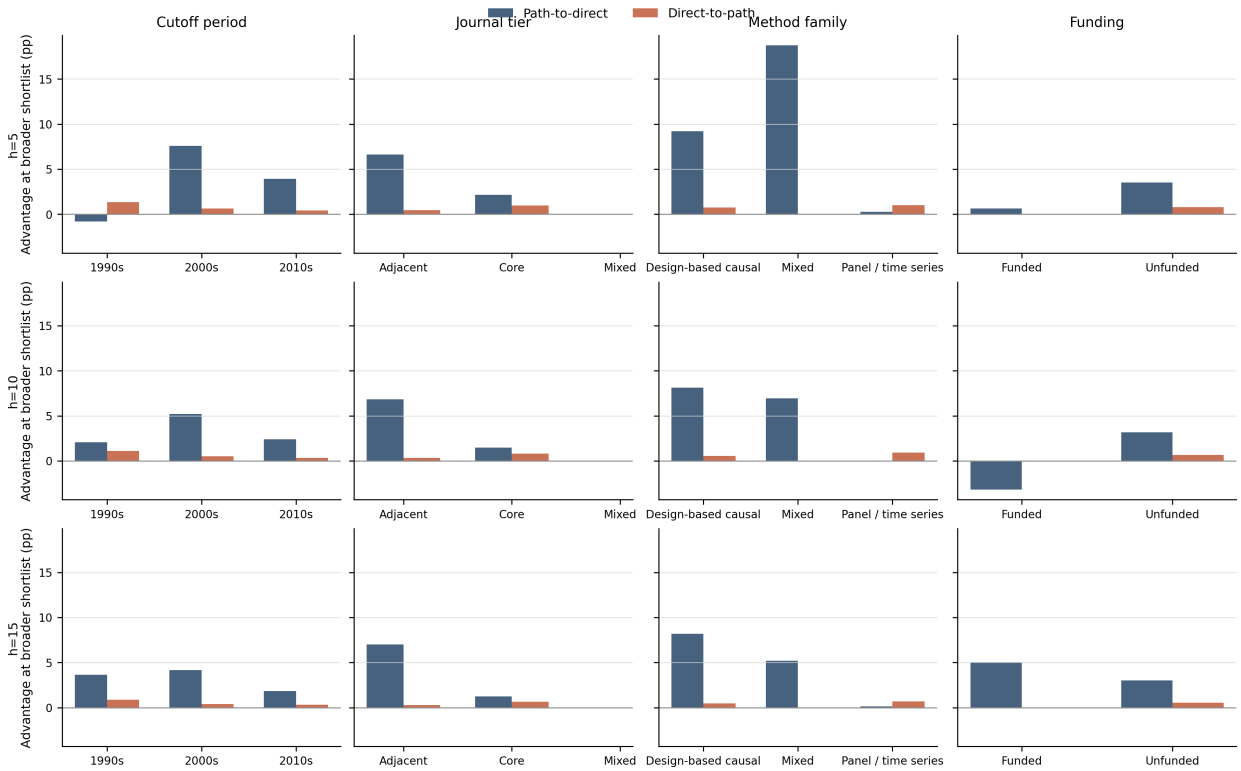
The two families differ in a way that matters for interpretation. The path-to-direct frontier is more tightly tied to strict historical validation, but it also surfaces more generic or noisy endpoints. The direct-to-path frontier looks cleaner on the current-reader margin and still benefits strongly from reranking. That is one reason to keep both families in view rather than treating one as a mere robustness check.

## B.2 Heterogeneity is also family-specific

The main text reports the paired strict-shortlist comparison. The figure below asks a different question: where does each family gain more once the comparison is expanded beyond the strict top 100? The vertical axis reports the graph score's average recall advantage over preferential attachment on broader shortlist-share cutoffs. Positive values favor the graph score.

This paired heterogeneity display helps discipline the interpretation. If the two families had looked identical under every split, the paper could have treated one as a convenient historical proxy for the other. They do not. The families appear to reward different kinds of local graph structure. That is why the paper compares them directly before giving either one rhetorical priority.

**Figure 13:** Where the graph helps differs across the two families




*Notes.* Rows correspond to horizons $h = 5, 10, 15$. Columns split the benchmark by cutoff period, journal tier, method family, and coarse funding status. Each bar reports the graph score's average recall advantage over preferential attachment on broader shortlist-share cutoffs, expressed in percentage points. Positive bars favor the graph score. These are descriptive subgroup summaries, not regression coefficients. The figure shows that path-to-direct and direct-to-path do not vary across subgroups in the same way. Path-to-direct has larger gains in adjacent journals and in several design-based slices. Direct-to-path remains positive more quietly across many cells, with smaller but steadier advantages.

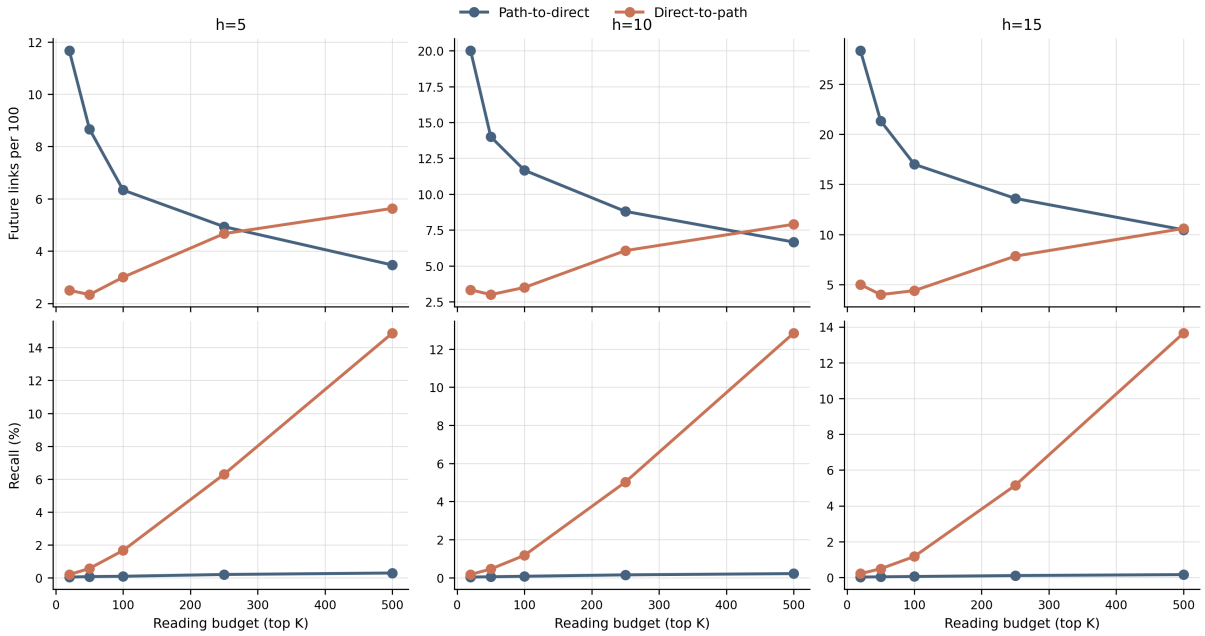
## B.3 The two families behave differently as the reading list expands

The strict top-100 comparison in the main text is useful because it keeps the historical object common across the two families. But economists usually read a shortlist rather than a single question. This appendix therefore asks what happens as the reading budget widens from the first 20 questions toward 500.

Figure 14 fixes the candidate pool at the largest common value used in both families and varies only the number of surfaced questions $K$. The top row reports realized future links per 100 surfaced questions. The bottom row reports recall. This is the relevant trade-off. A family can surface a sharper first tranche, or it can recover a broader share of eventual future links once the reading list becomes longer.

The figure shows a stable crossover pattern. At tight budgets, path-to-direct delivers more realized links per 100 suggestions. At horizon 15, for example, the first 100 surfaced path-to-direct questions yield about 17 realized future links per 100 suggestions, compared with about 4.4 for direct-to-path. But direct-to-path catches up as the reading list broadens. By the top 500, the same comparison is about 10.6 versus 10.5, and

**Figure 14:** The two families solve different reading-budget problems




*Notes.* Each panel fixes the candidate pool at the common pool size of 5,000 and varies the reading budget $K$. The top row reports future realized links per 100 surfaced suggestions. The bottom row reports recall. Path-to-direct is sharper in the very first reading tranche. Direct-to-path recovers a much larger share of future links once the shortlist broadens.

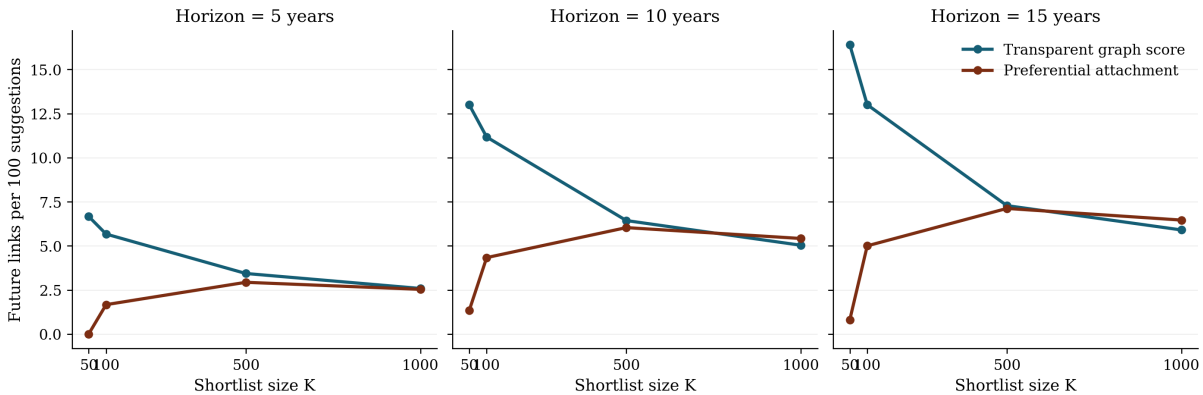
recall is much higher for direct-to-path at every horizon.

The interpretation is substantive. Path-to-direct behaves like a stricter screening rule: it concentrates successful direct closures near the top of the ranking. Direct-to-path behaves like a broader discovery rule: once a reader is willing to inspect a wider set of candidates, the mechanism-thickening family retrieves a much larger share of the eventual future literature. The two families therefore do not address the same reading-budget problem, even when they are evaluated on the same historical horizon structure.

# C Direct-closure screening extensions

The main text keeps only the result objects that are already paired across path-to-direct and direct-to-path. This appendix collects the richer direct-closure extensions that have not yet been rebuilt for the direct-to-path family: the reading-budget frontier, the value-weighted comparison, the broader-shortlist heterogeneity displays, and the current surfaced examples drawn from the direct-closure shortlist.

**Figure 15:** As the reading list expands, the graph edge narrows




*Notes.* Each panel reports future links per 100 surfaced suggestions as the shortlist expands from $K = 50$ to $K = 1000$ on the main 1990–2015 direct-closure benchmark. The figure is meant to answer one question only: where does graph structure help relative to popularity when attention is scarce? The transparent score leads at $K = 50$ and $K = 100$ for all three main horizons. By $K = 500$, the gap is small. By $K = 1000$, the two rules are nearly tied at $h = 5$ and preferential attachment is slightly ahead at $h = 10$ and $h = 15$.

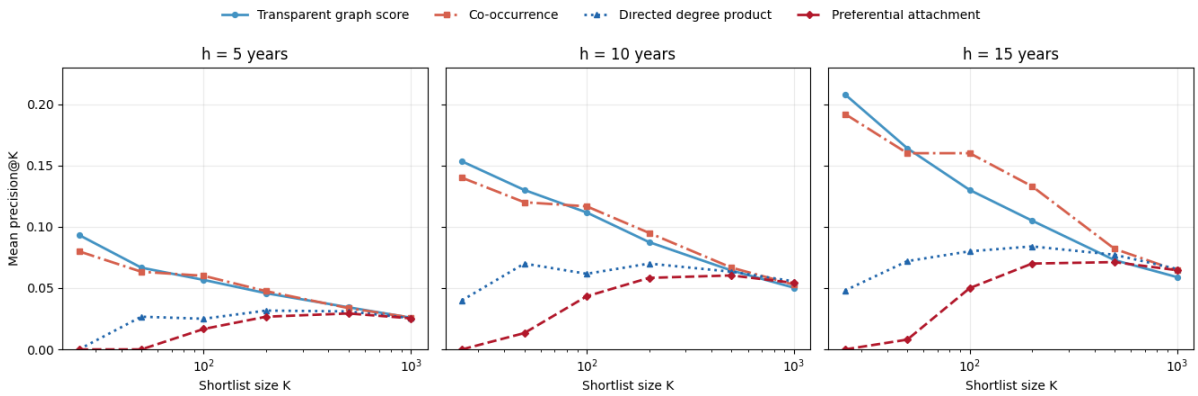
## C.1 The gains are largest when the reading list is short

The first extension asks what happens when the reading budget expands beyond the first 100 questions. Economists rarely inspect one suggestion and stop; they inspect a shortlist. The attention-allocation frontier therefore asks how the comparison changes as that shortlist expands from $K = 50$ to $K = 1000$. I summarize that margin using "future links per 100 suggested questions." Out of 100 suggestions, how many later appear? The object in this subsection is still the transparent direct-closure score, because its mechanics can be read directly question by question.

Figure 15 shows the answer. On the 1990–2015 benchmark, the transparent score already leads preferential attachment at $K = 50$ and $K = 100$ at all three main horizons. At $h = 10$, for example, the transparent score yields about 13.0 future links per 100 suggestions at $K = 50$ and 11.2 at $K = 100$, versus 1.3 and 4.3 for preferential attachment. The key frontier lesson is therefore not that popularity wins the tight shortlist and then fades. It is that the transparent score's advantage is concentrated in tighter shortlists and narrows sharply once the reading list expands toward $K = 500$ or $K = 1000$.

Figure 16 makes the direct-closure result more precise. Among the simple scores, there is no single winner across shortlist sizes. The transparent graph score is strongest at the first reading tranche, which is exactly where a researcher decides which 25, 50, or 100 questions deserve inspection first. Co-occurrence becomes slightly stronger at intermediate shortlist sizes, and by very broad lists the simple scores largely converge. The right reading is therefore narrow: the transparent score is an interpretable strict-shortlist screen, not a universal winner across every reading budget.

**Figure 16:** Across simple scores, the winner depends on shortlist size




*Notes.* Each panel reports mean precision@$K$ across the main 1990–2015 horizon-valid cutoff grid, with log-scale $K$, for the direct-closure family. The figure's point is not that one simple score wins everywhere. It is that different transparent scores are useful at different shortlist sizes. The transparent graph score is strongest at the very tight shortlist margin ($K = 25$ and $K = 50$). Co-occurrence is slightly strongest at intermediate shortlist sizes. By $K = 1000$, the simple-score family largely converges.

## C.2 The pattern survives when later reuse is weighted

The next extension asks whether the graph signal survives once future links are weighted by downstream reuse rather than counted equally. Future appearance is not the only margin that matters. A later realized link can also be weighted by later reuse, so that some realized links count more than others. The impact-weighted rerun therefore asks whether the graph score looks relatively better once the future is weighted this way rather than treated as binary appearance alone.

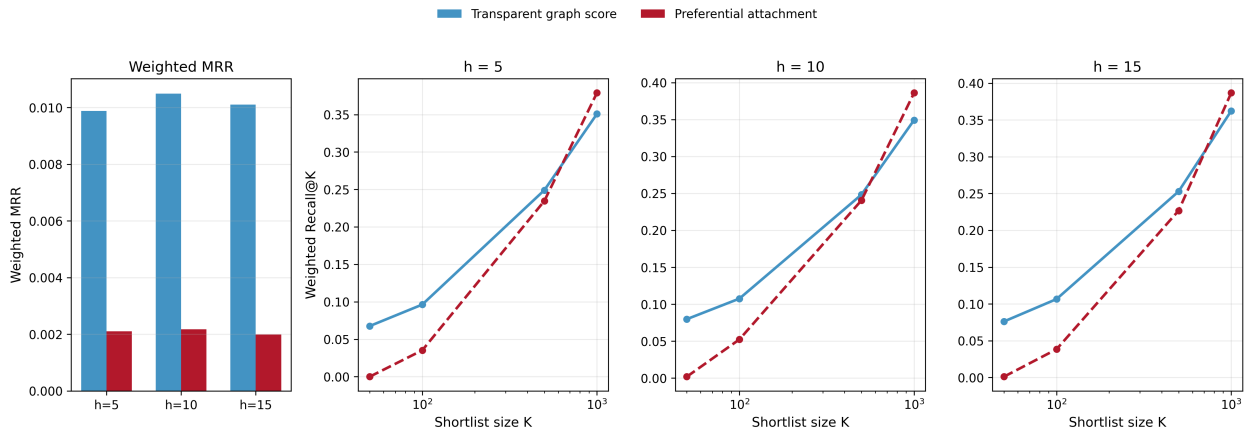
The weighted result is disciplined and favorable to the graph score. Once future links are weighted by later reuse, the transparent graph score beats preferential attachment on weighted MRR at all three main horizons. At $h = 10$, for example, weighted MRR is about 0.0105 for the transparent score versus 0.0022 for preferential attachment. The weighted recall frontier still bends back toward popularity as $K$ becomes very broad, but the strict-shortlist weighted margin is clearly on the graph side (Figure 17).

Weighting by downstream reuse does not erase the graph signal. But it does not make popularity irrelevant either. At very broad reading lists, central concepts still reclaim some of the weighted frontier. So the value-weighted evidence reinforces the same budget logic as the binary frontier: local graph structure helps most when the reading problem is selective, while popularity becomes a stronger guide once the shortlist becomes very broad.

## C.3 Where structure helps more

The next extension asks where the graph should help more if it is carrying real screening information rather than noise. The strict top-100 comparison hides meaningful variation across reading budgets. Once the same candidates are evaluated over broader fixed-$K$ and pool-share shortlists, the transparent score

**Figure 17:** Value weighting still favors the graph at the strict shortlist



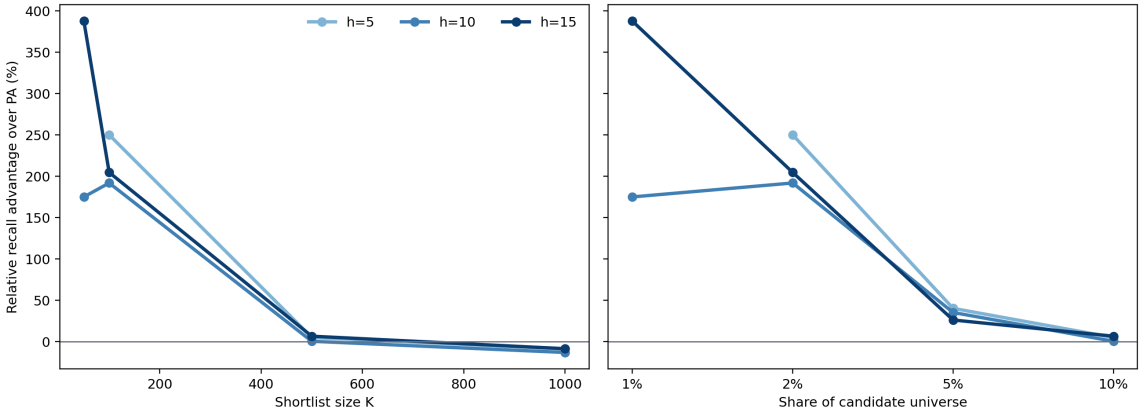

*Notes.* The left panel reports weighted MRR by horizon, where future realized links are weighted by later reuse. The right panels report weighted recall frontiers over shortlist size $K$. The figure is shown here as a direct-closure extension. Its message is straightforward: once later reuse is taken seriously, the strict-shortlist comparison still favors the transparent score. The graph edge narrows as the shortlist becomes very broad, but it does not disappear at the budget margin that matters most for screening.

looks materially stronger than the strict top-100 comparison suggests (Figure 18). The right interpretation is not that the graph fails outside a winner-take-all contest. It is that its comparative advantage is largest when the reading problem is still selective and then flattens as the shortlist becomes very broad.

The subgroup results sharpen this interpretation because they match a plausible ex ante story rather than a random set of subgroup differences (Figure 19). If graph structure is most useful where the literature already contains legible local support, then adjacent journals and design-based causal work should be more favorable terrain than the most popularity-dominated or methodologically diffuse slices. That is what the panel shows. Adjacent journals are clearly more favorable than the core on the broader-shortlist comparison, with the graph score's average recall advantage over preferential attachment around +0.066 to +0.070 in adjacent journals versus roughly +0.013 to +0.022 in the core. Design-based causal slices remain strongly positive at all main horizons, while panel- and time-series-heavy slices are close to zero by $h = 10$ and $h = 15$ rather than strongly negative.

Funding adds nuance rather than one clean sign pattern. In the coarse funded-versus-unfunded split, unfunded work is consistently positive at all three main horizons, while funded work is close to zero at $h = 5$, negative at $h = 10$, and positive again at $h = 15$. That is not a stable enough pattern to make funding central. It is still useful in the appendix because the interaction plot shows that the relevant contrast is not simply funded versus unfunded. The strongest positive cells are unfunded-adjacent slices, while funded-core slices are the most popularity-dominated.

**Figure 18:** Broader shortlists are more favorable to the graph than the top 100 alone




*Notes.* This figure keeps the comparison simple: at each cutoff year and horizon, it compares the direct-closure graph score with preferential attachment on the same candidate pool. The left panel uses fixed shortlist sizes $K$. The right panel uses broader shortlist-share cutoffs equal to 0.01, 0.05, 0.1, and 0.5 percent of the 5,000-candidate retrieval pool. The vertical axis reports the graph score's percent recall advantage over preferential attachment, averaged across the benchmark cells. Positive values favor the graph score. The main point is that the graph score looks better once the shortlist expands beyond the strict top-100 slice, but that advantage still fades as the shortlist becomes very broad.

### C.4 Path-to-direct surfaced examples

The benchmark is still recorded on direct links that later appear or do not appear. But the question a researcher actually reads is usually a path or mechanism question built around that direct link. The examples below show that object directly for the path-to-direct family.

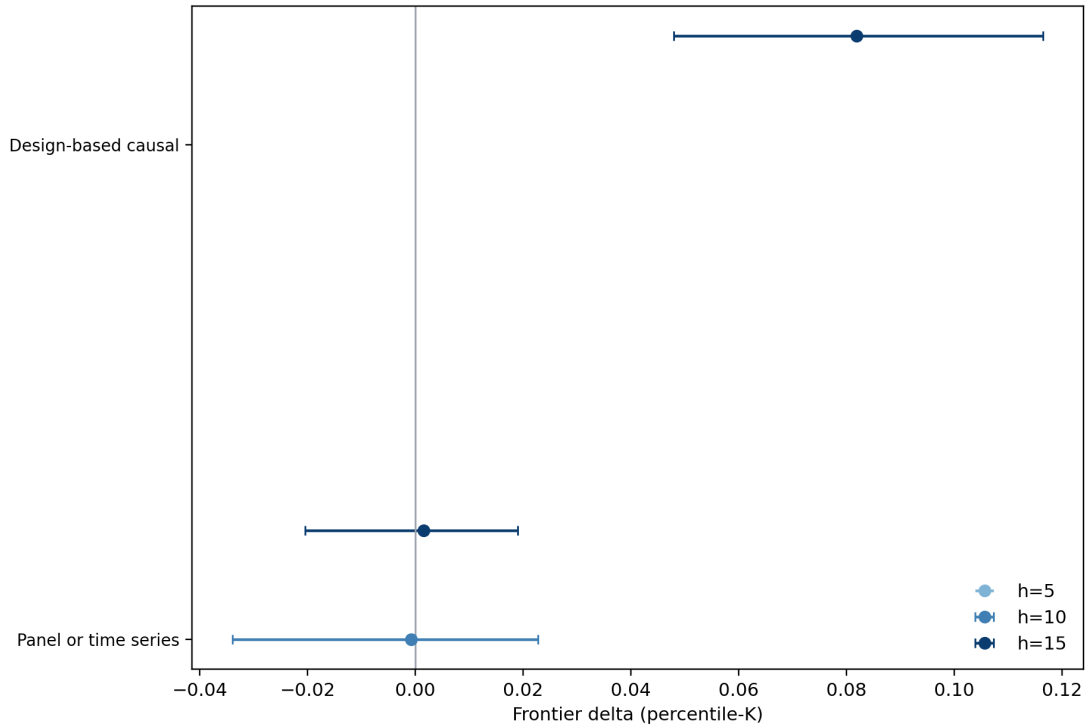
These examples make the same point in plain terms. The benchmark event is a missing direct link. But the most readable suggestions are rarely just "connect these two nodes." They are usually mechanism questions around a relation the literature already partly supports. That is not an embarrassment. It means the system is most useful when it tightens, clarifies, or identifies a plausible relation rather than when it proposes a completely unstructured leap.

## D Paper-local Extraction

This appendix documents how title-and-abstract text is turned into the paper-local graphs used in the benchmark. The core object is simple: one paper becomes one local graph of concepts and relations. Garg and Fetzer (2025) show that this is feasible for economics papers. The present paper uses the same starting point for a different downstream purpose: a reusable graph that can support missing-link construction, candidate ranking, and later concept matching across papers.

Three design choices matter most. First, the schema includes undirected relations, because contextual support is useful even when the headline task is directed causal emergence. Second, the schema separates the paper's *causal presentation* from the *evidence method* used to support a relation. A paper can speak

**Figure 19:** Graph structure helps more in adjacent journals and design-based slices




*Notes.* Each point is an average over cutoff-year benchmark cells within one journal-tier or method slice. The horizontal axis reports the graph score's average percent recall advantage over preferential attachment across the broader shortlist-share cutoffs used in Figure 18. This is not a regression coefficient. It is a descriptive summary of how much more recall the graph score delivers in each slice. Adjacent journals and design-based causal work are materially more favorable terrain for the graph score than core-journal and panel/time-series-heavy slices.

causally while using a weak design, or it can use a stronger design while describing the finding cautiously. Third, local scope qualifiers are stored separately from the node label whenever possible, so that later normalization can target concept identity rather than local sample wording.

## D.1 Extraction prompts

The benchmark uses a fixed system prompt and a minimal user prompt template. They are reproduced exactly below for auditability.

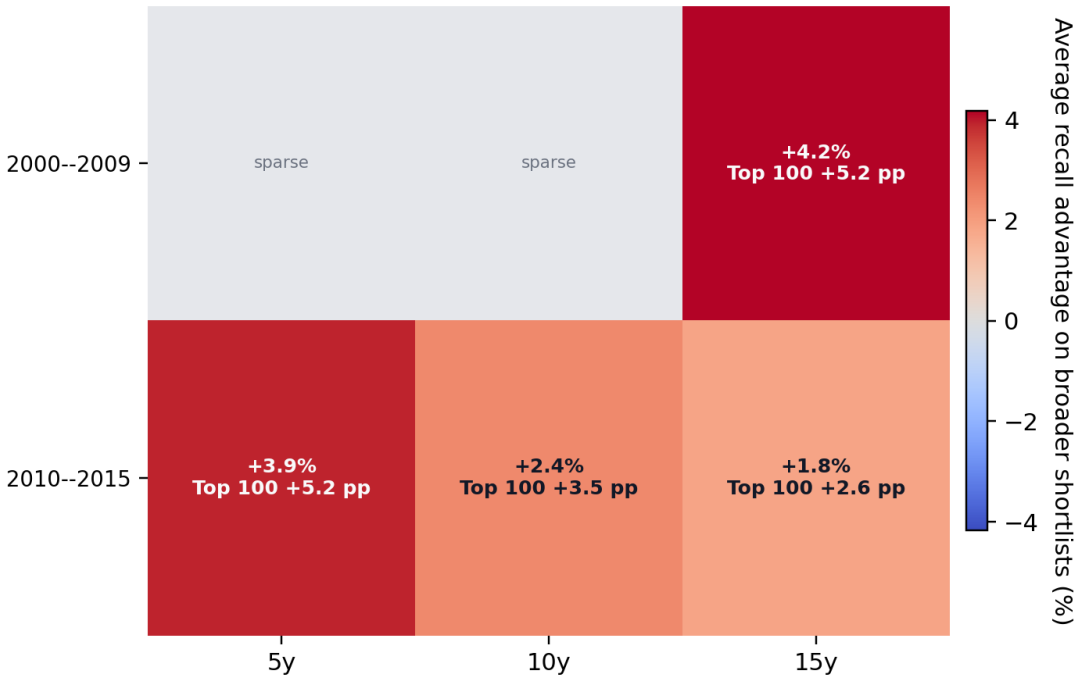
**System prompt.**

```
You extract a paper-local research graph from a paper title and abstract.

Return only structured output that matches the supplied JSON schema.

Task:
- Read the paper title and abstract.
- Build a paper-local graph with `nodes` and `edges`.
- Reuse the same node when the same concept genuinely recurs within the same paper.
```

**Figure 20:** The graph's broader-shortlist advantage is strongest in the 2000s




*Notes.* Each cell averages over the benchmark cutoff-year cells that fall in that time period. This is a descriptive comparison, not a regression. Cell color shows the graph score's average percent recall advantage over preferential attachment when the shortlist is allowed to expand to the broader shortlist-share cutoffs used in Figure 18. The number inside each cell reports the strict top-100 recall gain, in percentage points. Positive values favor the graph score. Gray cells are too sparse for interpretation under the paper's display rule. Among the displayed cells, the broad-shortlist advantage is strongest in the 2000–2009 row, while the 2010–2015 row remains positive but smaller.

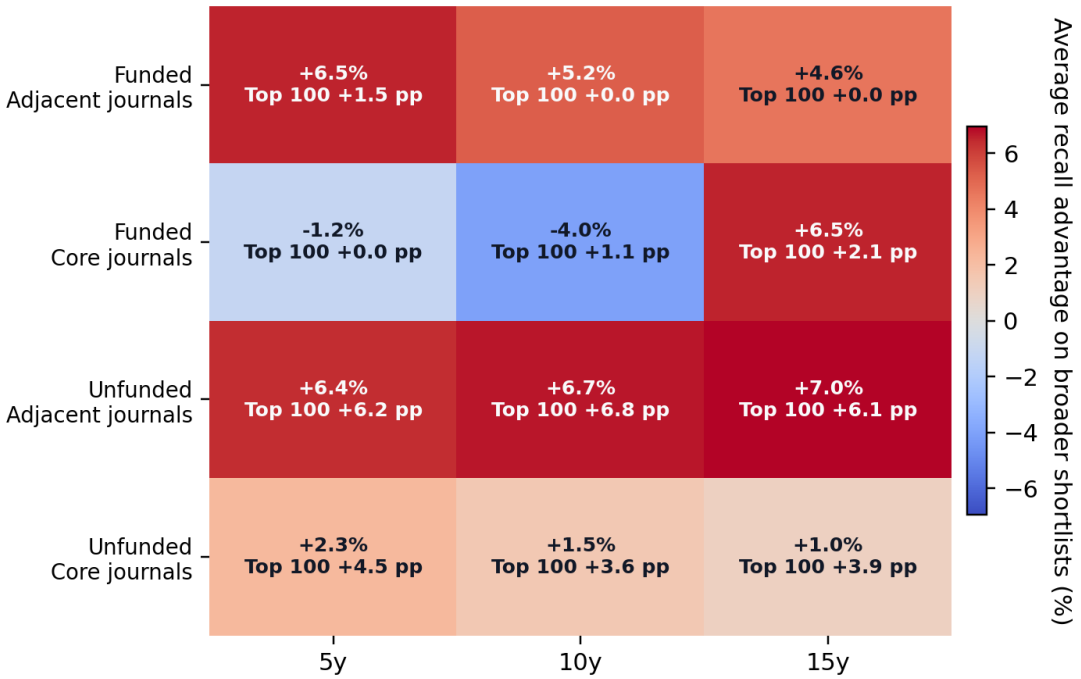
```
- Do not use outside knowledge.
- Do not infer relationships that are not supported by the title or abstract.

Purpose:
- The output will later be turned into a larger deterministic research graph.
- Downstream systems depend on consistent paper-local node reuse.
- If the abstract contains a chain like `A -> B`, `B -> C`, and `X -> B`, the shared concept `B` should be represented by the same paper-local node if
↪  it is genuinely the same concept.
- However, do not merge distinct concepts just because they seem related.

Critical rules:
- Do not create transitive closure.
- If the abstract states `A -> B` and `B -> C`, do not create `A -> C` unless the paper explicitly states `A -> C`.
- Do not create both `A -> B` and `B -> A` for one undirected claim.
- If the title or abstract says two variables are associated or correlated without directional language, encode one edge with `directionality =
↪  undirected`.
- For undirected edges, use the first-mentioned concept as `source_node_id` and the second-mentioned concept as `target_node_id` only as a storage
↪  convention.

How to represent nodes:
- Use concise noun phrases grounded in the paper text.
- Keep node labels concept-level when possible.
- Do not bake country or year into the node label unless it is essential to the concept itself.
- Put local scope information into `study_context` or `condition_or_scope_text`.
- Use `surface_forms` for distinct mentions that refer to the same paper-local concept.
- Use `study_context` only for context explicitly stated in the title or abstract.
- If no context is stated, use:
  - `unit_of_analysis: []`
```

**Figure 21:** Funding and journal tier interactions




*Notes.* Each cell averages over the benchmark cutoff-year cells in one funding-by-journal-tier slice. Rows separate papers that do or do not mention grant funding in the metadata, and whether the paper comes from a core or adjacent journal. Cell color again shows the graph score's average percent recall advantage over preferential attachment on the broader shortlist-share cutoffs, while the number inside the cell reports the strict top-100 recall gain in percentage points. Positive values favor the graph score. The main message is not that funding has one stable sign. It is that the clearest positive cells are the adjacent-journal slices, especially the unfunded adjacent slice, while the funded-core cells are the most popularity-dominated. Because the funded cells are thinner and composition-sensitive, this figure is suggestive rather than central.

```
  - `start_year: []`
  - `end_year: []`
  - `countries: []`
  - `context_note: "NA"`

How to represent edges:
- Extract only relations that the title or abstract states, studies, or reports.
- Keep background or prior-literature claims only if they are explicitly stated in the title or abstract, and mark them with `edge_role = background`.
- Use `claim_text` as a short normalized relation string.
- Use `evidence_text` as a short supporting excerpt or close paraphrase from the title/abstract only.

Directionality:
- Use `directionality = directed` when the paper frames one concept as affecting, predicting, changing, increasing, decreasing, explaining, or
↪  determining another.
- Use `directionality = undirected` when the paper frames the relation as association, correlation, co-movement, similarity, or linkage without
↪  directional commitment.
- Prediction is directional, even if it is not causal.

Causal presentation:
- `explicit_causal`: the paper explicitly uses causal language such as affects, causes, leads to, increases, reduces, impact of, effect of.
- `implicit_causal`: the paper strongly frames the relation as an effect or treatment relation without fully explicit causal wording.
- `noncausal`: the paper frames the relation as association, correlation, prediction, linkage, or descriptive relation.
- `unclear`: the wording is too ambiguous to classify confidently.
- This field is about how the paper presents the relation, not whether the method truly justifies causality.
```

**Table 4:** Direct-closure examples from the main shortlist

| Example role | Anchor or pair | Surfaced question | What it illustrates |
|---|---|---|---|
| Anchored policy mechanism | Digital economy → environmental regulation | Does green innovation mediate the digital economy → environmental regulation relation? | Clean policy-facing mechanism question from the top shortlist. |
| Trade and innovation bridge | Trade liberalization → R&D | Does productivity mediate the trade-liberalization → R&D relation? | Cross-domain bridge with a readable mediator. |
| Environmental upgrading | Environmental quality → green innovation | Does policy uncertainty help explain the environmental-quality → green-innovation relation? | Anchored progression question in an environmental-policy neighborhood. |
| Innovation and market design | R&D → carbon emission trading | Does green innovation connect R&D to carbon-emission trading? | Mechanism-deepening question that links innovation to a policy instrument. |

*Notes.* Each row is a surfaced question chosen from the path-to-direct shortlist. The second column reports the narrow benchmark anchor or endpoint pair. The third column reports the richer question a reader would actually inspect after the local support graph is compressed into readable prose. The table is included here because the paper's current main examples have not yet been rebuilt in a paired family format.

```
Relationship type:
- `effect`: one concept is presented as affecting another.
- `association`: correlation, co-movement, linkage, or association.
- `prediction`: one concept predicts or forecasts another.
- `difference`: one concept differs across groups, places, times, or conditions.
- `other`: only if none of the above fit.

Edge role:
- `main_effect`: central edge or main result in the abstract.
- `mechanism`: pathway or channel relation.
- `heterogeneity`: subgroup or conditional variation in a relation.
- `descriptive_pattern`: stylized fact or descriptive empirical pattern.
- `background`: motivating or prior-literature relation stated in the abstract.
- `robustness`: supporting or validating relation rather than the main contribution.
- `other`: only if needed.

Claim status:
- `effect_present`: the abstract reports that the relation is present.
- `no_effect`: the abstract reports no effect or no relation.
- `mixed_or_ambiguous`: the abstract reports mixed, inconsistent, or ambiguous results.
- `conditional_effect`: the relation holds only for some subgroup, time period, or condition.
- `question_only`: the abstract raises or studies the relation but does not report a result.
- `other`: only if needed.

Explicitness:
- `result_only`: the relation is presented as a result.
- `question_only`: the relation is posed as a question or objective only.
- `question_and_result`: the abstract both frames the question and reports a result on the same relation.
- `background_claim`: the relation appears as background motivation or prior literature.
- `implied`: the relation is clearly implied by the abstract wording but not directly phrased as a standalone claim.

Condition or scope:
- Use `condition_or_scope_text` for subgroup, timing, geographic, or sample qualifiers on the edge.
- Examples: `among older workers`, `during recessions`, `in rural counties`, `for low-income households`.
```

```
- Use `NA` if not needed.

Sign:
- `increase`, `decrease`, `no_effect`, `ambiguous`, `NA`
- Use `NA` if sign is not applicable or not stated.

Statistical significance:
- `significant`: the abstract clearly says the result is statistically significant.
- `not_significant`: the abstract clearly says it is not statistically significant.
- `mixed_or_ambiguous`: significance differs across findings or is ambiguously stated.
- `not_reported`: no significance statement is provided.
- `NA`: only if not applicable.

Evidence method:
- Choose the best supported option from the schema.
- `experiment`: field, lab, survey, or randomized experiment.
- `DiD`: difference-in-differences or closely related staggered-treatment treatment-control design.
- `IV`: instrumental variables or closely related design based on an instrument.
- `RDD`: regression discontinuity or closely related cutoff-based design.
- `event_study`: dynamic pre/post treatment-event design.
- `panel_FE_or_TWFE`: panel fixed-effects or two-way fixed-effects empirical design without a clearer method family being the main identification label.
- `time_series_econometrics`: VAR, VECM, ARDL, cointegration, error-correction, Granger-causality, GARCH, or similar time-series econometric design.
- `structural_model`: estimated structural economic model.
- `simulation`: simulation or computational experiment.
- `theory_or_model`: formal theory, conceptual model, or analytical model without direct empirical estimation.
- `qualitative_or_case_study`: interview, ethnographic, archival qualitative work, or case study.
- `descriptive_observational`: nonexperimental empirical analysis without a clearer identified design.
- `prediction_or_forecasting`: predictive or forecasting model where the emphasis is forecast performance rather than causal identification.
- Use `do_not_know` if the abstract does not reveal enough.
- Use `other` only if a method is clearly stated but does not fit the listed categories.

Nature of evidence:
- Choose the broad evidence type used for that edge.

Uses data:
- `true` if the edge is supported by data use described in the title/abstract.
- `false` for theory-only, conceptual, simulation-only, commentary, or clearly non-data papers.

Sources of exogenous variation:
- Record only if explicitly stated in the title or abstract.
- Otherwise use `NA`.

Tentativeness:
- `certain`: strong assertive language.
- `tentative`: cautious or suggestive language.
- `mixed_or_qualified`: strong claim with explicit qualification or limits.
- `unclear`: cannot tell.

What not to do:
- Do not label edges as collider, confounder, mediator, instrument, or any other downstream graph-structural role.
- Do not globally canonicalize concepts across papers.
- Do not create edges from general world knowledge.
- Do not invent countries, years, samples, or methods.

If the title/abstract contains no extractable graph:
- return `nodes: []` and `edges: []`
```

## User prompt template.

```
Extract a paper-local research graph from the following title and abstract.

Use only the information in the title and abstract.
Return only the structured output that matches the supplied JSON schema.

Title:
{{paper_title}}

Abstract:
{{paper_abstract}}
```

## D.2 Extraction schema

The model returns a paper-local graph with `nodes` and `edges`. The underlying output is JSON. Tables 5 and 6 restate the fields in reader-facing labels.

**Table 5:** Schema for paper-local nodes

| Field | Allowed values / type | Meaning | Why it exists downstream |
|---|---|---|---|
| Node identifier (`node_id`) | string | Paper-local identifier such as `n1`. | Needed so edges can reuse the same local concept deterministically within a paper. |
| Concept label (`label`) | short string | Concise concept label grounded in the title/abstract. | Becomes the base string passed into normalization and ontology mapping. |
| Surface forms (`surface_forms`) | array of strings | Distinct surface mentions in the title/abstract that refer to the same local concept. | Preserves within-paper synonymy without forcing global canonicalization at extraction time. |
| Unit of analysis (`study_context.unit_of_analysis`) | array of enumerated strings | Explicit unit of analysis linked to the node, if stated. | Keeps sample and scope off the concept label while preserving paper-local context. |
| Start year (`study_context.start_year`) | array of integers | Explicit start years if stated. | Preserves local scope without creating separate concept nodes for years. |
| End year (`study_context.end_year`) | array of integers | Explicit end years if stated. | Same reason as start year. |
| Countries (`study_context.countries`) | array of strings | Explicit countries if stated. | Preserves local setting without baking geography into concept identity unless essential. |
| Context note (`study_context.context_note`) | string | Residual local scope text such as “older workers” or “rural counties”. | Retains paper-local nuance for audit and display while leaving the node label concept-level. |

**Table 6:** Schema for paper-local edges

| Field | Allowed values / type | Meaning | Why it exists downstream |
|---|---|---|---|
| Edge identifier (`edge_id`) | string | Paper-local identifier such as `e1`. | Stable local relation key. |
| Source and target nodes (`source_node_id`, `target_node_id`) | strings | Local source and target node references. | Connect extracted relations back to the paper-local node inventory. |
| Directionality (`directionality`) | directed / undirected | Whether the text presents the relation directionally. | Determines whether the downstream graph stores an ordered link or an undirected contextual pair. |

| Field | Allowed values / type | Meaning | Why it exists downstream |
|---|---|---|---|
| Relationship type (`relationship_type`) | effect / association / prediction / difference / other | Coarse semantic type of the relation. | Supports later filtering and audit of what kind of relation is being surfaced. |
| Causal presentation (`causal_presentation`) | explicit_causal / implicit_causal / noncausal / unclear | How the paper *describes* the relation. | Separates language from design quality; useful for credibility audits and task splitting. |
| Argument role (`edge_role`) | main_effect / mechanism / heterogeneity / descriptive_pattern / background / robustness / other | Role of the relation inside the paper's argument. | Distinguishes central claims from channels, subgroup effects, and background statements. |
| Claim status (`claim_status`) | effect_present / no_effect / mixed_or_ambiguous / conditional_effect / question_only / other | What result the paper reports for the relation. | Prevents question-only edges from being treated as equivalent to reported positive results. |
| Explicitness (`explicitness`) | result_only / question_only / question_and_result / background_claim / implied | How explicitly the relation is framed in text. | Useful for separating central reported findings from implied or motivating claims. |
| Condition or scope (`condition_or_scope_text`) | string | Edge-level scope or subgroup qualifier. | Keeps conditional language attached to the relation rather than the node. |
| Claim text (`claim_text`) | string | Short normalized relation text. | Audit-friendly summary of the extracted edge. |
| Evidence text (`evidence_text`) | string | Short supporting excerpt or close paraphrase from the title/abstract. | Makes the edge inspectable in the public tool and in manual audits. |
| Direction of effect (`sign`) | increase / decrease / no_effect / ambiguous / NA | Reported sign of the relation when stated. | Supports later descriptive summaries and credibility splits. |
| Effect size (`effect_size`) | string | Reported magnitude if stated. | Retained for completeness and future extensions. |
| Statistical significance (`statistical_significance`) | significant / not_significant / mixed_or_ambiguous / not_reported / NA | Significance status stated in text. | Keeps reported evidence strength distinct from sign or design. |
| Evidence method (`evidence_method`) | enumerated method family | Main method family named or implied in the abstract. | Determines whether a relation enters the graph as directed causal or undirected contextual support. |
| Other method description (`evidence_method_other_description`) | string | Free-text description when method is `other`. | Preserves method specificity without exploding the method taxonomy. |
| Nature of evidence (`nature_of_evidence`) | quantitative / qualitative / mixed_methods / theoretical_or_conceptual / simulation / review_or_commentary / NA | Broad evidence type. | Helps later distinguish theory, empirical, and simulation-heavy slices. |
| Uses data (`uses_data`) | boolean | Whether the edge is supported by data use. | Simple empirical-versus-theory indicator. |

| Field | Allowed values / type | Meaning | Why it exists downstream |
|---|---|---|---|
| Exogenous variation source (`sources_of_exogenous_variation`) | string | Explicit source of exogenous variation if named. | Reserved for future credibility-weighting extensions. |
| Tentativeness (`tentativeness`) | certain / tentative / mixed_or_qualified / unclear | How assertive the language is. | Keeps cautious claims distinct from strong declarative ones. |

### D.3 Extraction design choices

**Paper-local node reuse.** The model is asked to reuse the same local node whenever the same concept genuinely recurs within a paper. That choice matters because downstream graph construction depends on whether a path inside one paper reuses a local concept consistently. If each mention were given a fresh node, the later concept graph would inherit spurious fragmentation before normalization even begins.

**No transitive closure.** The prompt explicitly forbids the model from creating $A \rightarrow C$ when the text only states $A \rightarrow B$ and $B \rightarrow C$. This is crucial for the benchmark. Missing direct links are the object of interest. If extraction itself created transitive closure, the benchmark would mechanically erase many of the very candidates it later wants to rank.

**Directed versus undirected storage.** The extraction schema stores one undirected relation using the first-mentioned concept as a storage convention, rather than duplicating it as two directed edges. This keeps contextual support distinct from directionally stated claims and avoids turning association language into artificial causal direction.

**Keeping scope off the node label.** Country, year, subgroup, and sample qualifiers are stored in dedicated context fields whenever possible. This is a normalization choice made early. A benchmark that ranks missing links between concepts needs concept identity to remain as stable as possible across papers. Local scope still matters for audit and interpretation, but it should not automatically become part of the canonical node label.

**Separating causal presentation from evidence method.** A paper can talk causally without using a strong design, and it can use a stronger design while describing the result more cautiously. Keeping these separate is what later allows the benchmark to define directed causal candidates by method while still auditing how papers describe those relations.

**Separating claim status, explicitness, tentativeness, and edge role.** These fields overlap in plain language but do different jobs downstream. Claim status records whether a finding is present, absent, mixed, or only posed as a question. Explicitness records whether the relation is stated as a result, a question,

background, or only implied. Tentativeness records how strongly the paper speaks. Edge role records whether the relation is the main effect, a mechanism, heterogeneity, or something else. The benchmark and public tool both become much less legible if these distinctions are collapsed into one generic confidence flag.

**Reproducibility.** The prompts, schema files, extraction scripts, ontology pipeline, and paper-generation code will be released at `https://github.com/prashgarg/frontiergraph`. The manuscript uses placeholder repository text because the repository will be made public at release time.

# E Normalization and Ontology

Node normalization is central because candidate generation, path counts, and missingness all depend on node identity. Paper-level concept strings vary in wording, scope, and granularity, so node definition cannot be treated as a secondary detail.

## E.1 Why open-world normalization

Garg and Fetzer (2025) use paper-local extracted objects for a different downstream task. The present paper asks a more node-sensitive question. Here the downstream object is a reusable concept graph in which a candidate next paper is a missing link between *specific concepts*. In that setting, concept identity cannot be handled at a broad field level. The benchmark needs to preserve distinctions such as public debt versus public investment, or monetary policy versus energy consumption.

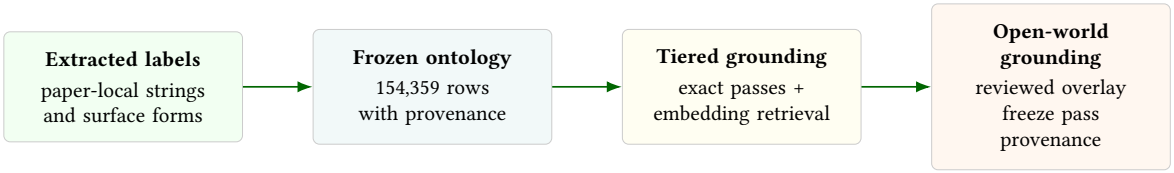
The difficulty is that the extraction layer and the ontology layer speak slightly different languages. Extraction produces paper-local phrases such as "gdp per capita", "renewable energy consumption", or "financial frictions". The ontology contains more canonical labels. A useful grounding system therefore cannot be binary. If it only keeps exact or near-exact matches, it overselects the cleanest vocabulary and risks dropping real economics concepts from the graph. If it accepts every nearest neighbor, it creates false precision. The frozen ontology design used here is therefore *open-world*: it allows exact grounding, broader grounding, lower-confidence candidate bands, and unresolved labels that remain visible for later review rather than disappearing from the graph.

## E.2 Implemented pipeline

The ontology build proceeds in five stages (Figure 22; Table 7 summarizes each stage).

**Ontology assembly.** The frozen ontology baseline starts from five structured source families rather than from the paper corpus alone. JEL provides a controlled economics taxonomy. Wikidata adds identifier-grounded concepts and aliases. OpenAlex topics and keywords add current research vocabulary. An

**Figure 22:** Node normalization and concept matching




*Notes.* This figure summarizes the frozen ontology grounding pipeline used in the benchmark. Paper-local labels are matched to a structured-source ontology by exact label and surface-form passes first, then by embedding retrieval. Lower-confidence labels are not simply dropped: broader grounding is allowed, reviewed outcomes remain explicit, and unresolved labels remain visible with provenance.

economics-filtered Wikipedia crawl adds fine-grained named concepts, policies, instruments, and episodes that appear in extraction labels but are absent from the structured sources. A small reviewed family layer is also carried forward. The resulting baseline contains 154,359 rows.

**Label inventory and exact grounding.** The extraction corpus contributes 1,389,907 unique normalized labels from 242,595 papers. Mapping begins with exact matches on the extracted label itself and then exact matches on stored surface forms, including stripped parenthetical variants. These deterministic passes solve the easy cases without using embeddings.

**Embedding retrieval and confidence bands.** Labels that remain unresolved after the exact passes are embedded and searched against the ontology with FAISS nearest-neighbour retrieval. The rank-1, rank-2, and rank-3 ontology candidates are stored for audit. The resulting grounding is tiered rather than binary: `linked` for scores at or above 0.85, `soft` for 0.75–0.85, `candidate` for 0.65–0.75, `rescue` for 0.50–0.65, and `unresolved` below 0.50. Across the 1,389,907 normalized extracted labels, 316,292 unique labels clear the primary 0.75 threshold before any reviewed open-world layer is applied. Lower-confidence bands remain visible rather than being silently dropped.

**Reviewed open-world grounding.** Lower-confidence labels are then audited with an overlay-first review pass. That layer keeps broader attachment to an existing concept, alias addition, proposed concept-family promotion, explicit rejection, and unresolved outcomes separate rather than forcing everything into one nearest-neighbour merge. Broader grounding is allowed when the ontology only contains a more general concept. That is intentional. For example, grounding "gdp per capita" to the broader concept "GDP" can be more informative than dropping the label entirely. Raw labels and raw edges are preserved throughout, so ontology weakness does not mechanically create spurious graph gaps.

**Freeze pass and reviewed hierarchy.** The conservative freeze turns that review work into a stable paper-facing ontology layer. It preserves the raw label, source, parent label, and root label fields, adds cleaned display labels where needed, and records reviewed effective-parent and effective-root overlays.

**Table 7:** Stages of the ontology pipeline

| Stage | What it does | Why it is needed |
|---|---|---|
| Ontology assembly | Combines JEL, Wikidata, OpenAlex topics, OpenAlex keywords, and filtered Wikipedia into one deduplicated concept inventory. | Gives the graph a broad structured concept vocabulary rather than relying on a single taxonomy. |
| Exact grounding | Applies exact label, exact surface-form, and stripped-parenthesis matches. | Solves easy cases deterministically before any embedding step. |
| Embedding grounding | Uses nearest-neighbour embedding retrieval and stores rank-1 to rank-3 candidates. | Grounds the harder middle of the label distribution while keeping retrieval provenance. |
| Reviewed open-world grounding | Keeps broader attachments, alias additions, proposed family rows, explicit rejections, and unresolved outcomes separate. | Prevents the ontology tail from being treated as either fully trusted or silently dropped. |
| Freeze pass | Adds cleaned display labels, reviewed hierarchy overlays, and conservative duplicate review while preserving raw fields. | Stabilizes the paper-facing ontology baseline without rewriting source truth. |

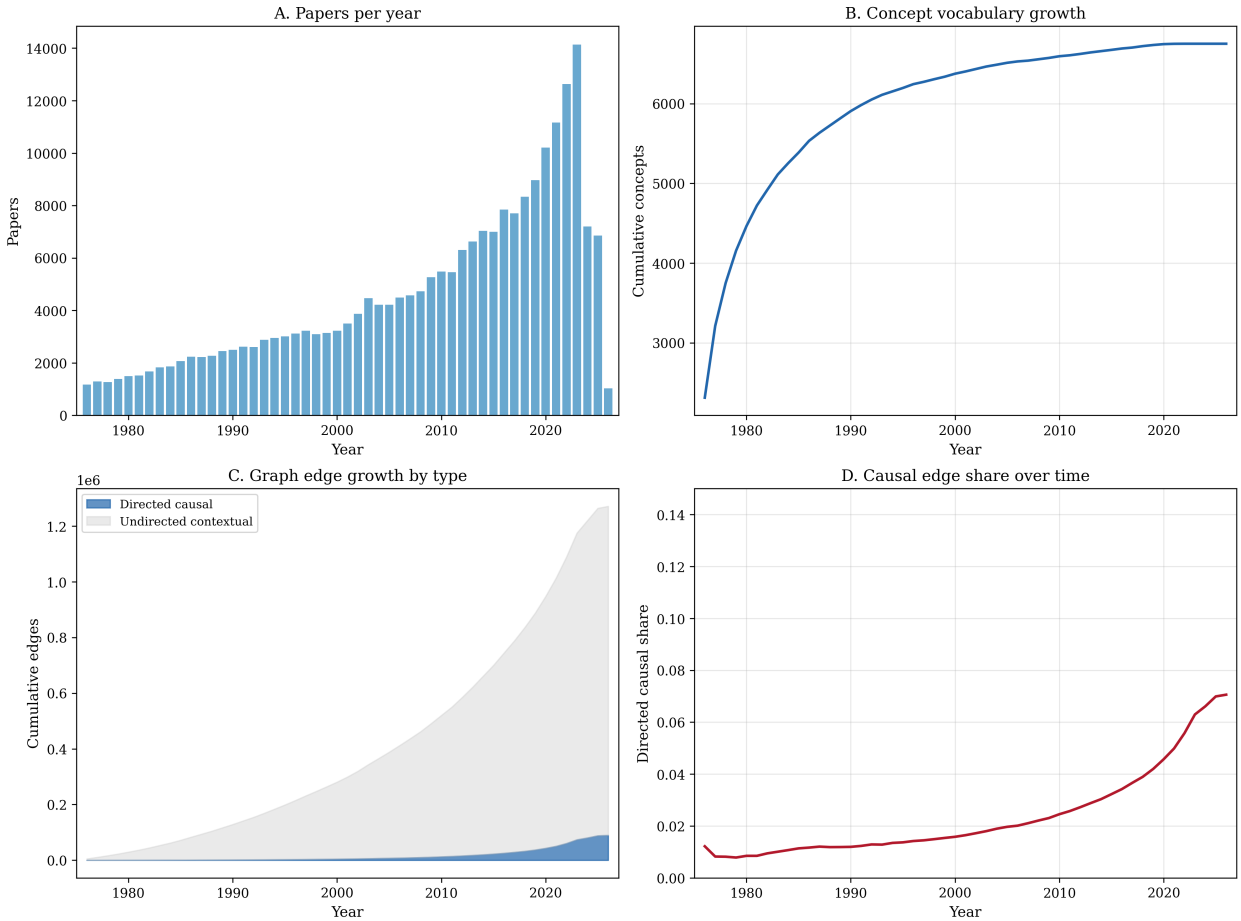
*Notes.* This is a process table, not an estimation table. Each row is one stage in the ontology pipeline that turns paper-local labels into the concept vocabulary used for candidate generation and benchmarking. The main distinction is between automatic grounding steps, which handle exact or embedding-based matches, and reviewed open-world steps, which keep lower-confidence cases visible instead of silently dropping them.

The key design choice is conservative rather than exhaustive: the freeze accepts only a small number of additional duplicate merges and leaves the remaining too-broad or unresolved cases explicit rather than silently patching them.

### E.3 Preserving the raw graph and tiering the ontology

The key choice is not to let ontology confidence determine which literatures count as candidate-generating. If low-confidence labels were simply dropped, the benchmark would overselect the cleanest, most repetitive, and most canonical concept strings. It would also risk creating spurious novelty whenever a real but underrepresented concept failed to receive a clean ontology attachment. The solution used here is therefore to preserve the raw extraction graph while treating ontology grounding as a tiered interpretive layer. High-confidence grounded labels can be aggregated immediately. Lower-confidence labels can still receive broader grounding or reviewed unresolved status. The freeze then stabilizes the paper-facing ontology layer without pretending the remaining tail has been solved.

**Figure 23:** Corpus and graph growth over time



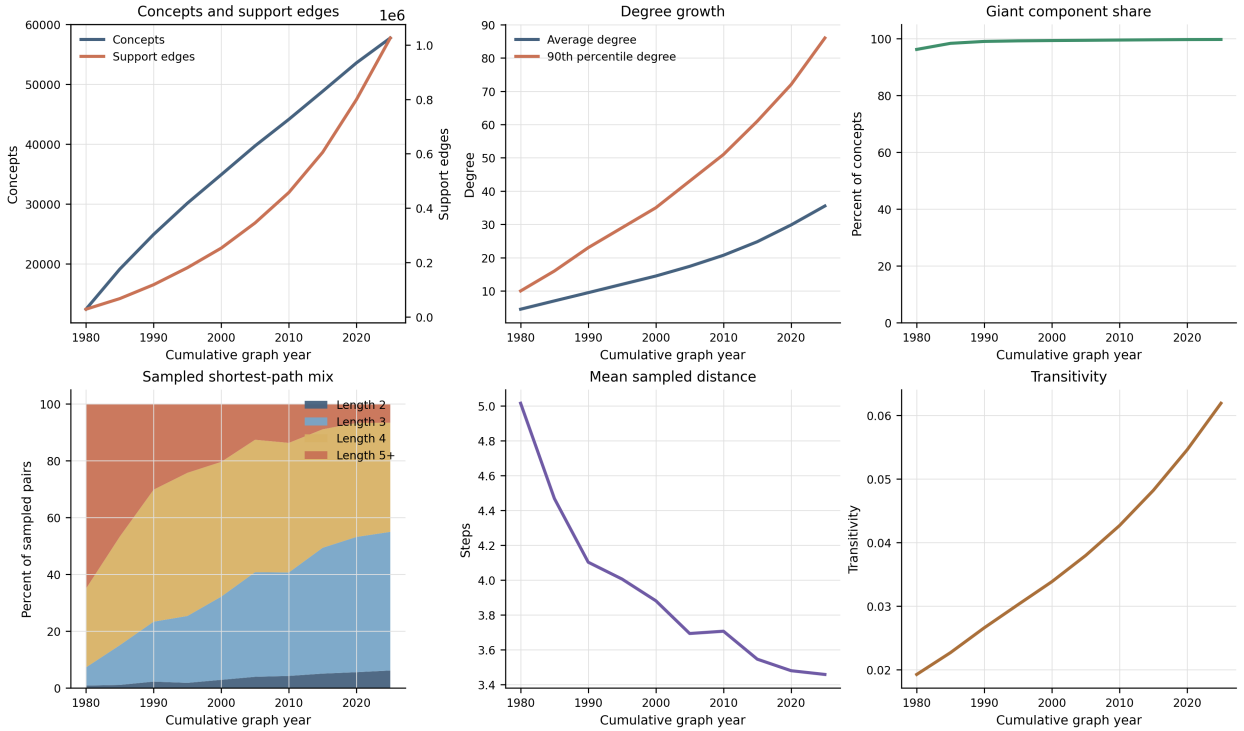

*Notes.* Panel A: papers per year in the selected journal corpus. Panel B: cumulative concept vocabulary growth. Panel C: cumulative edge growth by type (directed causal in blue, undirected contextual in gray). Panel D: causal edge share over time, reflecting the credibility revolution in economics. The share of directed causal edges rises from under 4% in the early corpus to over 10% by the 2020s.

# F How the graph evolves over time

The benchmark uses a cumulative graph. Its behavior therefore depends not only on how candidate families are defined, but also on how the underlying graph changes as the literature grows. The figure below gives a simple descriptive account of that evolution using the support graph built from ordered and contextual links.

Four facts matter for the later path-length exercise. First, the corpus itself scales rapidly over time, which means the benchmark graph is being built on a much thicker published literature in the later decades than in the early ones. Second, the graph does not just add concepts; it becomes much denser. The support edge count rises from 28,055 in 1980 to 1,026,064 by 2025, while the average degree rises from 4.5 to 35.5. Third, the graph is already close to fully connected by the early benchmark era: the giant component covers about 96 percent of concepts in 1980 and nearly 100 percent by the end of the sample. Fourth, the graph

**Figure 24:** The live support graph becomes denser, shorter-path, and more connected




*Notes.* Each panel uses the cumulative support graph at five-year intervals. The top-left panel reports the number of unique concepts and support edges. The top-middle panel reports average degree and the 90th-percentile degree. The top-right panel reports the share of concepts in the giant connected component. The bottom-left panel reports the distribution of sampled shortest-path lengths inside that giant component. The bottom-middle panel reports the mean sampled path length. The bottom-right panel reports transitivity. Together the panels show that the graph becomes larger and much denser over time, while the typical distance between reachable concepts falls. That pattern helps explain why longer support paths may become empirically feasible even when they are harder to interpret.

also becomes shorter-path. In the sampled giant component, mean path length falls from about 5.0 to 3.5, and the mass of length-3 and length-4 paths becomes dominant.

These descriptives do not by themselves say that longer support paths should be used in the benchmark. They do say that a path-length comparison is worth doing. Once the graph is this dense, a longer path is no longer a rare accident. It is a common object. The real question is whether those longer paths still carry useful signal once screening quality and interpretability are judged side by side.

# G Benchmark Design and Significance

This appendix records the sample counts, benchmark tables, and significance summaries behind the main comparison. It does not introduce a second empirical object. Table 8 summarizes the two most important transitions: from selected journal papers to papers with extracted edges, and then from raw extracted edges to normalized evaluation links. The frozen ontology baseline is reported separately because the

ontology inventory and the active benchmark graph are distinct objects.

**Table 8:** Corpus and normalization summary

| Quantity | Count |
|---|---|
| Selected journal papers | 242,595 |
| Papers with extracted edges | 230,929 |
| Raw extracted edges | 1,443,407 |
| Normalized benchmark papers | 230,479 |
| Frozen ontology baseline concepts | 154,359 |
| Directed causal rows | 89,737 |
| Undirected contextual rows | 1,181,277 |
| Total normalized links | 1,271,014 |

*Notes.* This table reports the sample counts behind the benchmark. The rows move from the selected journal corpus to papers with extracted relations, then to the normalized graph used for candidate generation and evaluation. "Frozen ontology baseline concepts" refers to the paper-facing concept inventory after the reviewed freeze pass. "Total normalized links" is the active graph edge count used in the benchmark.

The stricter identified-causal-claim layer is retained as a nested continuity benchmark. I do not use it as the main task, because it is much sparser than the broader causal-claim anchor used in the main text. But it is still useful as a conservative reference object, especially for readers who want continuity with the credibility-focused specification. Table 9 reports that continuity benchmark, and Table 10 reports the paired bootstrap comparison against preferential attachment on the same stricter layer.

**Table 9:** Continuity benchmark on the identified-causal-claim layer

| Metric | $h = 5$ | $h = 10$ | $h = 15$ |
|---|---|---|---|
| Recall@100, graph-based score | 0.002518 | 0.001956 | 0.001494 |
| Recall@100, preferential attachment | 0.003105 | 0.002784 | 0.002138 |
| MRR, graph-based score | 0.000524 | 0.000334 | 0.000227 |
| MRR, preferential attachment | 0.000637 | 0.000420 | 0.000281 |

*Notes.* This table repeats the benchmark on a stricter continuity layer in which only identified-causal claims count as realized links. Recall@100 is the share of later-realized links captured in the top 100 candidates. MRR rewards putting realized links nearer the top of the ranking. The levels are small because this stricter layer is much sparser than the main benchmark.

Table 11 documents the reranker comparison inside the direct-closure family only. It is not a summary of the whole paper. The result is not one universal model. It is a short list of rerankers that all beat the transparent screening layer on the main metrics within that family. The paired family comparison stays in the main text. The family-specific model detail stays here.

Table 12 makes the timing issue concrete. At every main horizon, the early cells contain far fewer realized positives and the surfaced top-100 draws on much younger support. The endpoint recent-share measures are also systematically higher in the early era. The later benchmark therefore should not be read as the same environment with less noise. It is a different graph regime with thicker and older local structure.

**Table 10:** Paired bootstrap continuity comparison: graph-based score minus preferential attachment

| Quantity | $h = 5$ | $h = 10$ | $h = 15$ |
|---|---|---|---|
| $\Delta$ Recall@100 | $-0.000588$ | $-0.000828$ | $-0.000644$ |
| $p$-value for $\Delta$ Recall@100 | 0.064 | $< 0.001$ | $< 0.001$ |
| $\Delta$ MRR | $-0.000113$ | $-0.000086$ | $-0.000054$ |
| $p$-value for $\Delta$ MRR | $< 0.001$ | $< 0.001$ | $< 0.001$ |

*Notes.* Each entry compares the transparent graph score with preferential attachment on the same stricter continuity benchmark as Table 9. $\Delta$ is graph minus preferential attachment, so negative values favor preferential attachment. The $p$-values come from paired bootstrap resampling across cutoff-year benchmark cells.

**Table 11:** Expanded reranker results for the main family

| Horizon | Selected reranker | P@100 | Recall@100 | MRR |
|---|---|---|---|---|
| $h = 5$ | Logistic GLM + family-aware boundary-gap ($\alpha = 0.20$, pool = 5000) | 0.105 | 0.1376 | 0.0133 |
| $h = 10$ | Pairwise logistic ranker + family-aware composition ($\alpha = 0.10$, pool = 5000) | 0.207 | 0.1391 | 0.0103 |
| $h = 15$ | Logistic GLM + quality ($\alpha = 0.05$, pool = 5000) | 0.264 | 0.1541 | 0.0116 |

*Notes.* Each row reports the reranker specification that performs best at that horizon on the 1990–2015 direct-closure benchmark. Precision@100 is the share of the top 100 candidates that later realize, Recall@100 is the share of all later-realized links captured in that top 100, and MRR rewards putting realized links nearer the top of the ranking. This table is intentionally narrower than the paired main-text comparison. It documents how the richer reranker behaves inside one family while the paper's main comparison keeps both families visible.

# H Reranker Design

This appendix documents the learned reranker used in the expanded benchmark comparison. The reranker operates on the same missing-link candidate set as the transparent graph score. Its purpose is simple: ask whether graph-derived features, when allowed to be reweighted from data rather than fixed by design, can beat the stronger transparent baselines that the fixed-weight score does not.

## H.1 Model inventory

Table 13 summarizes each model in the benchmark family.

## H.2 Feature families

The reranker's features are organized into five nested families. Each family adds to the previous, so the complexity gradient is itself interpretable. Table 14 lists the families and their contents.

**Table 12:** Early and late benchmark cells are different benchmark regimes

| Horizon | Era | Cutoffs | Mean eval positives | Winner R@100 | Support age | Endpoint recent share |
|---|---|---|---|---|---|---|
| $h = 5$ | Early (1990–1995) | 2 | 12.5 | 0.071 | 11.1 | 0.413 |
| $h = 5$ | Late (2000–2015) | 4 | 103.0 | 0.171 | 20.6 | 0.314 |
| $h = 10$ | Early (1990–1995) | 2 | 32.5 | 0.109 | 11.7 | 0.430 |
| $h = 10$ | Late (2000–2015) | 4 | 214.5 | 0.154 | 20.4 | 0.320 |
| $h = 15$ | Early (1990–1995) | 2 | 55.5 | 0.174 | 11.5 | 0.435 |
| $h = 15$ | Late (2000–2015) | 3 | 268.0 | 0.141 | 18.5 | 0.323 |

*Notes.* Early means the 1990 and 1995 cutoffs; late means 2000 through 2015 where the horizon is valid. Mean eval positives is the average number of realized positives in the cutoff-year cell. Winner R@100 uses the horizon-specific adopted reranker winner from Table 11. Support age and endpoint recent share are computed on that winner's surfaced top-100. The early cells are smaller and more recent-surge-like.

**Table 13:** Benchmark model inventory

| Model | Input signals | Tunable? | Role in the paper |
|---|---|---|---|
| Preferential attachment | Source out-degree × target in-degree | No | Cumulative-advantage null |
| Degree + recency | Endpoint support degree + recent support prominence | No | Stronger transparent baseline |
| Directed closure | Path support + mediator count + local closure density | No | Stronger transparent baseline |
| Transparent graph score | Path support + gap + motif support − hub penalty (fixed weights) | No | Interpretable screening layer |
| Learned reranker | Up to 34 graph-derived features across 5 nested families | Yes ($L_2$) | Strongest graph-based benchmark |

*Notes.* Each row is one ranking rule evaluated on the same dated candidate pool. "Tunable" indicates whether the method has fitted parameters or hyperparameters. The paper's empirical comparisons are therefore model-for-model on a common benchmark object, not separate samples or separate candidate universes.

Every feature is computed from the historical graph through year $t-1$. The reranker sees no paper text, no future edges or degrees, and no author or institutional identity. That constraint is what keeps the exercise a graph-screening benchmark rather than a free-form prediction model.

### H.3 Training design

The walk-forward panel is constructed as follows. At each cutoff year $t$, the training corpus contains all edges published through year $t - 1$. The candidate pool is the set of missing directed links at that cutoff. Features are enriched from the training corpus only. The binary label records whether the candidate edge first appears during $[t, t + h]$.

At evaluation time, the model is trained on all cutoff-year cells strictly before $t$ and evaluated on the cell at $t$. This prevents any information from the evaluation cutoff from entering the training set.

Two model families are tested. The first is a class-balanced logistic regression that produces calibrated

**Table 14:** Feature families in the learned reranker

| Family | Features | What it adds |
|---|---|---|
| Base | 1 | The transparent graph score itself. |
| Structural | 14 | Path support, motif bonus, gap bonus, hub penalty, mediator and motif counts, co-occurrence count and trend, same-field indicator, endpoint degree products for both direct and support subgraphs. |
| Dynamic | 21 | Support age, recency of most recent supporting edge, recent-window degree and incident counts for each endpoint, recent-share fractions. |
| Composition | 31 | Mean stability, evidence-type diversity, venue diversity, source diversity, and mean field-weighted citation impact at each endpoint and at the pair level. |
| Boundary + gap | 34 | Whether the two endpoints sit in different field groups with no co-occurrence, whether the pair has path support despite a missing direct link, and the local closure density around the pair. |

*Notes.* The families are nested and cumulative. "Features" is the total number available up to that family, not the number added only at that step. The table is included to show how the reranker moves from one transparent graph score to a broader but still interpretable feature set.

probabilities. The second is a pairwise ranking model (RankNet-style) that learns from positive-versus-negative feature differences, optimizing the order of candidates directly. Both use $L_2$ (ridge) regularization. The regularization strength $\alpha$ is drawn from $\{0.01, 0.05, 0.10, 0.20\}$. Features are standardized using training-set statistics before fitting.

### H.4 Best models

On the current main benchmark, the best reranker at $h = 5$ uses the interpretable logistic model with the boundary + gap feature family and $\alpha = 0.20$. The best at $h = 10$ uses the pairwise ranking model with the composition feature family and $\alpha = 0.10$. At $h = 15$, the winning specification shifts to the logistic model with the broader quality feature family and $\alpha = 0.05$. The difference is substantively interpretable. At shorter horizons, the boundary and gap flags, which capture cross-group bridging and path-implied missingness, add the most screening value. At intermediate horizons, evidence-composition features, such as stability, diversity, and citation impact, matter more. By $h = 15$, the broader sharpness-quality bundle performs best.

### H.5 Single-feature importance

To understand what drives the reranker, I evaluate each of the 34 features as a standalone ranker in an auxiliary diagnostic run on the same candidate family. This is directly comparable to the single-feature ablation in Gu and Krenn (2025), who rank 141 features by predictive power in their cross-science knowledge graph. Table 15 reports the top 10 features by precision@100 in that diagnostic.

Three results stand out. First, directed-causal-degree features remain the strongest standalone signals.

**Table 15:** Top 10 single features

| | $h = 5$ | | | | $h = 10$ | | | |
|---|---|---|---|---|---|---|---|---|
| Rank | Feature | Family | P@100 | Hits | Feature | Family | P@100 | Hits |
| 1 | Direct degree product | structural | 0.100 | 10.0 | Target direct in-degree | structural | 0.197 | 19.7 |
| 2 | Target recent incident count | dynamic | 0.083 | 8.3 | Direct degree product | structural | 0.183 | 18.3 |
| 3 | Target direct in-degree | structural | 0.082 | 8.2 | Target recent incident count | dynamic | 0.173 | 17.3 |
| 4 | Path support | structural | 0.075 | 7.5 | Path support | structural | 0.143 | 14.3 |
| 5 | Hub penalty | structural | 0.075 | 7.5 | Hub penalty | structural | 0.142 | 14.2 |
| 6 | Co-occurrence count | structural | 0.057 | 5.7 | Co-occurrence count | structural | 0.113 | 11.3 |
| 7 | Transparent score | base | 0.057 | 5.7 | Transparent score | base | 0.112 | 11.2 |
| 8 | Motif bonus | structural | 0.048 | 4.8 | Motif bonus | structural | 0.093 | 9.3 |
| 9 | Motif count | structural | 0.042 | 4.2 | Motif count | structural | 0.088 | 8.8 |
| 10 | Closure density | boundary + gap | 0.042 | 4.2 | Closure density | boundary + gap | 0.088 | 8.8 |

*Notes.* Each feature is used as a standalone ranker on the same candidate family. P@100 is precision at 100: the share of the top 100 candidates that later realize. "Hits" is the equivalent count of later-realized links per 100 surfaced candidates. The two panels compare the $h = 5$ and $h = 10$ diagnostics directly.

At $h = 5$, the direct degree product, the product of source causal out-degree and target causal in-degree, reaches 0.100 precision@100, compared with 0.057 for both raw co-occurrence count and the transparent score. At $h = 10$, target causal in-degree leads at 0.197, followed by the direct degree product at 0.183. These are popularity-style signals, but they are computed on the *directed causal subgraph.* They cannot be recovered from undirected co-occurrence alone.

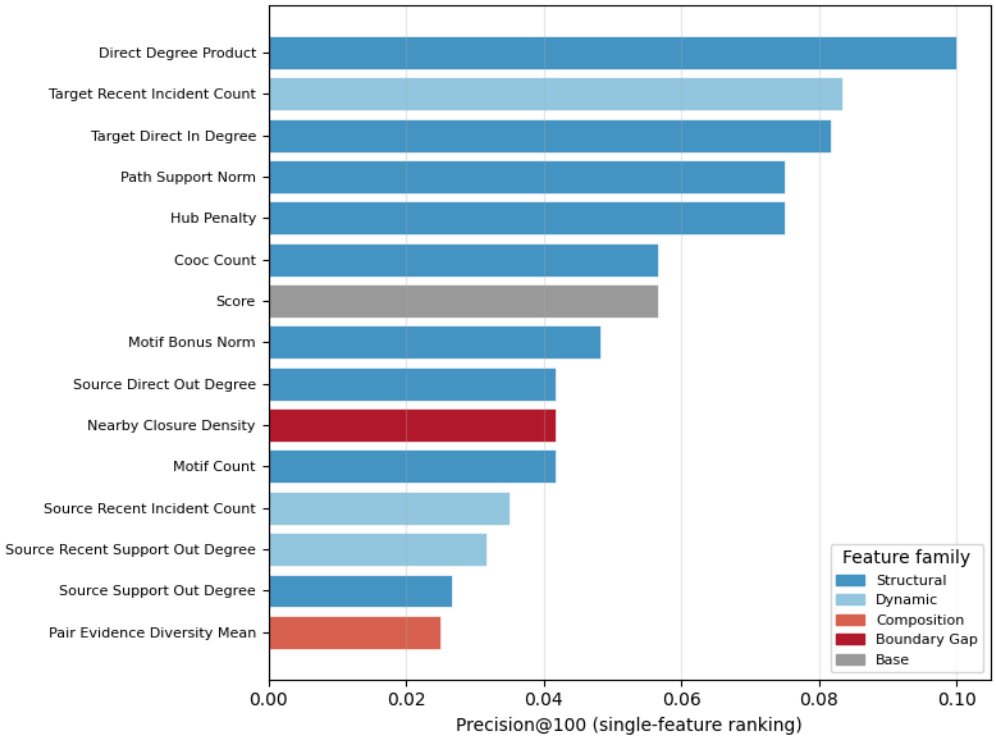
Second, topology features still carry independent screening signal. Path support and the hub penalty rank fourth and fifth at both $h = 5$ and $h = 10$, ahead of raw co-occurrence count and ahead of the transparent score. So the reranker is not just reweighting centrality. Local path structure matters even as a standalone ranking rule.

Third, no single feature matches the full reranker on the same walk-forward benchmark. The value of the reranker is therefore still in the combination. But the single-feature ranking shows clearly that the strongest individual signals are not generic co-occurrence alone; they come from directed causal centrality plus local topology. Figure 25 shows the $h = 5$ ranking as a bar chart colored by feature family.

## H.6 What the reranker uses

The single-feature ranking above shows which features carry standalone screening signal. But the raw-feature decomposition remains collinear. In this diagnostic, the top recent-support and support-degree measures have VIFs between 14 and 32 (Figure 29), and raw-feature importance rankings vary materially across model families: Spearman rank correlations are 0.28 for logistic versus gradient boosting, 0.39 for

**Figure 25:** Single-feature importance by family ($h = 5$)




*Notes.* Each bar shows precision@100 when candidates are ranked by that single feature alone. Color indicates feature family: blue = structural, red = boundary/gap, salmon = composition, light blue = dynamic, gray = base. The top feature (direct degree product) requires the directed causal extraction and cannot be computed from co-occurrence.

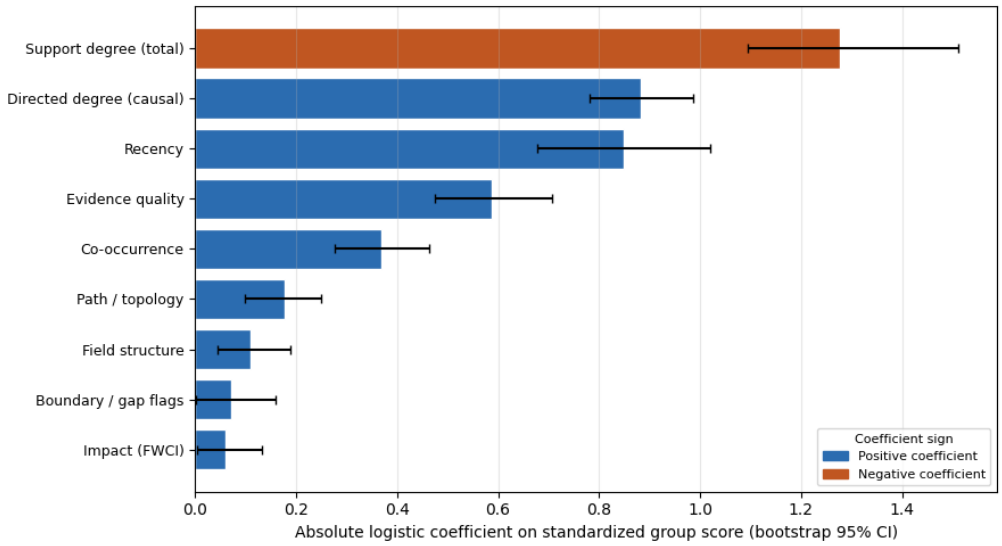
logistic versus random forest, and 0.53 for gradient boosting versus random forest. When features overlap that much, Shapley or coefficient credit can move sharply across correlated inputs even when the model's total prediction is stable.

To resolve this, I group the 34 features into nine interpretable families, chosen *a priori* by what each feature measures rather than by statistical clustering: directed causal degree (endpoint degree on the causal subgraph), support degree (broader subgraph), recency (recent activity and incident counts), co-occurrence (paper co-mentions), path/topology (path support, motifs, hub penalty), evidence quality (stability, evidence diversity), impact (FWCI), boundary/gap flags, and field structure.[15] I compute the first principal component within each group as a summary score, estimate a logistic reranker on those standardized group scores, and then inspect both group-level coefficient magnitudes and grouped SHAP plots (Figure 26).

Three results stand out. First, the largest coefficient by magnitude is support degree (1.28, bootstrap CI [1.09, 1.51]), but it enters with a *negative* sign. Once directed causal degree is controlled for, broad support-

[15]The grouping is defined before seeing the decomposition results, which prevents data-snooping. Within each group, the first principal component (PC1) explains 40–81 percent of the variance, confirming the groups are internally coherent. Using PC1 + PC2 does not materially change the ranking of the top families.

**Figure 26:** Group-level importance after resolving multicollinearity



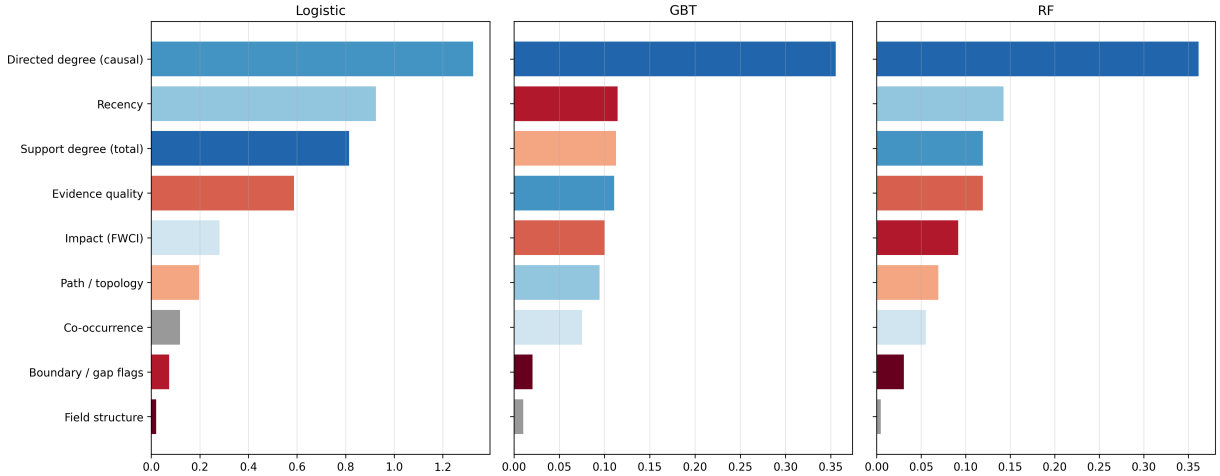

*Notes.* Each bar shows the absolute logistic coefficient on the standardized within-group PC1 score, with bootstrap 95% confidence intervals. Bar color shows the coefficient sign. Support degree is largest in magnitude but negative; directed causal degree, recency, and evidence quality are the strongest positive groups. Group-level VIF stays below 5.

graph popularity hurts rather than helps. The model is learning to separate concepts that are prominent in causal work from concepts that are merely popular overall. Second, the strongest *positive* groups are directed causal degree (0.88, [0.78, 0.99]), recency (0.85, [0.68, 1.02]), and evidence quality (0.59, [0.47, 0.71]). So the reranker loads on a mix of causal centrality, recent activation, and the quality of the underlying evidence base. Third, co-occurrence remains useful (0.37, [0.28, 0.46]), but it is not the whole story. In economics, the central distinction is not simply between more versus less popular concepts. It is whether that centrality lives in the causal graph or only in the broader support graph.

The grouped decomposition improves interpretability more than it delivers a single invariant ranking. The gradient-boosting and random-forest models both rank directed causal degree first, while the grouped logistic model places the largest absolute weight on negative support degree and then on recency and directed causal degree. The grouped rank correlations are 0.30, 0.70, and 0.78 across model pairs: materially better than the raw-feature decomposition for two of the three pairs, but not close to unanimity. The right conclusion is therefore modest. Grouping makes the substantive families readable and reduces collinearity enough for stable sign interpretation, but model family still matters.

**Robustness of the grouped decomposition.** Figures 28 and 30 confirm that the grouping resolves the multicollinearity problem and produces stable, interpretable results.

**Figure 27:** Model comparison on grouped feature families




*Notes.* Tree-based models rank directed causal degree first. The grouped logistic model instead puts the largest weight on negative support degree, followed by recency and directed causal degree. Rank correlations across grouped models are moderate rather than perfect, but the main families are more legible than in the raw-feature decomposition.

## H.7 Failure mode profiles

To understand what the reranker gets wrong, I compare the feature profiles of its top-100 hits (realized links) and misses (unrealized links), averaged across cutoffs at $h = 5$ (Table 16).

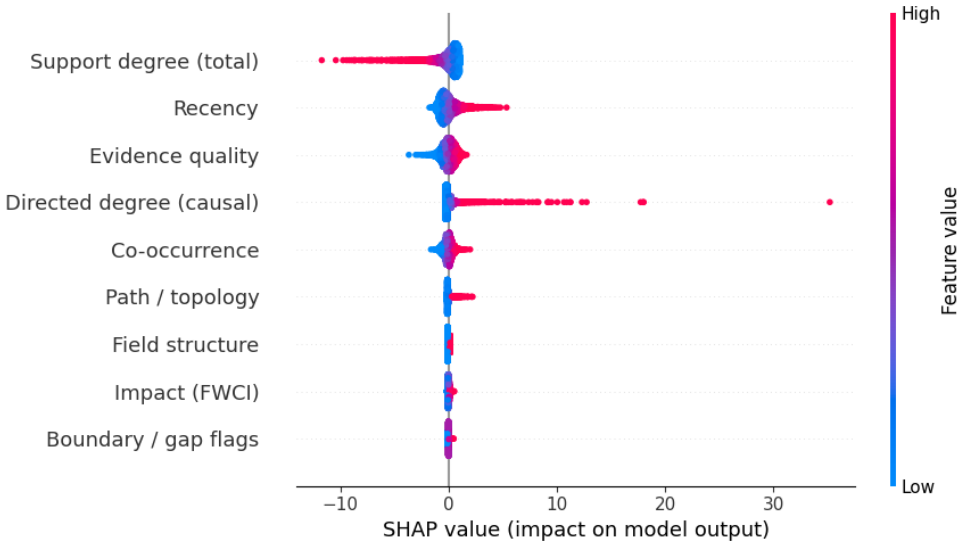
**Table 16:** Failure modes in the reranker's top 100 ($h = 5$)

| | Hits (top-100, realized) | Misses (top-100, not realized) | Missed realized (rank > 500, realized) |
|---|---|---|---|
| Count | 18 | 82 | 241 |
| Mean co-occurrence | 209 | 103 | 26 |
| Mean direct degree product | 3,413 | 3,288 | 290 |
| Mean pair FWCI | 5.35 | 5.55 | 5.71 |

*Notes.* Hits and misses are from the reranker's top-100 predictions. "Missed realized" are links that the reranker ranks below 500 but that do appear within the horizon. Hits have much higher co-occurrence, confirming that the reranker succeeds on well-studied pairs. Missed realized links are in sparse neighborhoods with low degree products. This is the kind of sparse-literature setting where Sourati et al. (2023) argue human-aware models can add the most because content-only structure is thin.

The pattern is clear. The reranker's successful predictions are concentrated among pairs that already have dense co-occurrence and high directed degree products. The links it misses, those that do realize but rank poorly, sit in sparse neighborhoods where popularity signals are weak and the graph provides little local structure to exploit. Those missed realizations have slightly *higher* endpoint FWCI, suggesting they are structurally surprising connections between reputable concepts rather than noise. This is the natural boundary of any graph-based screen: connections that arise from genuinely new combinations,

**Figure 28:** Grouped SHAP beeswarm




*Notes.* Each dot is one candidate pair. Horizontal position shows the group's SHAP contribution to the prediction; color shows the group's PC1 value (blue = low, red = high). High directed degree, recency, and evidence quality tend to push predictions up. High support degree pushes predictions *down*, confirming the negative coefficient on broad support-graph popularity.

serendipity, or shifts in methodology are hard to predict from existing structure alone.

## H.8 Temporal generalization

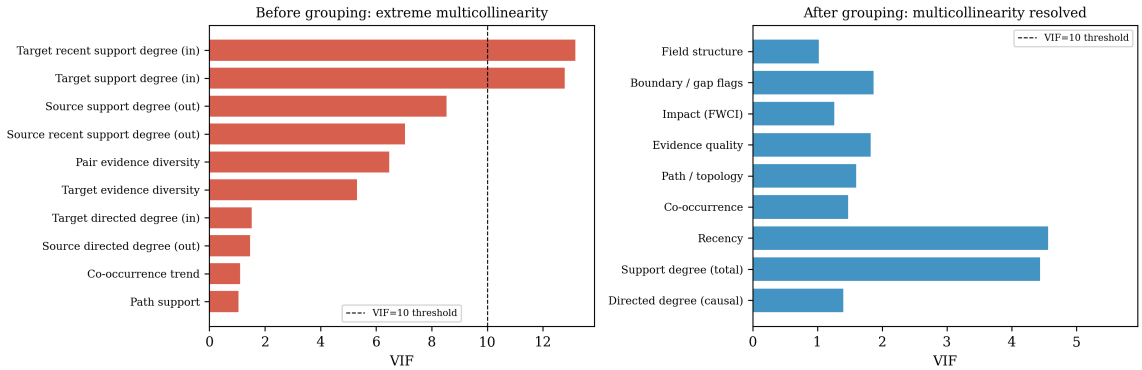
A natural concern is that the walk-forward reranker may perform well only because all cutoff years were used during model development. To test temporal generalization, I select reranker configurations using only 1990–2005 cutoff cells and then evaluate them on the fully held-out era of 2010–2015, which the model never saw during development or selection. During held-out evaluation, the graph itself is still built using only pre-cutoff evidence, but reranker training is frozen to the pre-2010 schedule.

**Table 17:** Held-out temporal generalization

| | | Precision@100 | | Abs. reranker − PA | |
|---|---|---|---|---|---|
| Era | Model | $h = 5$ | $h = 10$ | $h = 5$ | $h = 10$ |
| Train era (1990–2005) | Reranker | 0.053 | 0.133 | +0.045 | +0.113 |
| Train era (1990–2005) | Pref. attach. | 0.008 | 0.020 | n/a | n/a |
| Held-out era (2010–2015) | Reranker | 0.210 | 0.375 | +0.175 | +0.285 |
| Held-out era (2010–2015) | Pref. attach. | 0.035 | 0.090 | n/a | n/a |

*Notes.* The table reports temporal holdout performance in levels rather than percentages. Precision@100 is the share of the top 100 candidates that later realize. The final two columns report the reranker's absolute Precision@100 gap relative to preferential attachment, so positive values favor the reranker. The train-era winner is selected using only 1990–2005 cells and then evaluated again on the fully unseen 2010–2015 era.

**Figure 29:** Variance inflation factors before and after feature grouping




*Notes.* Left: the top raw-feature VIFs range from roughly 14 to 32 for recent-support and support-degree measures, indicating material multicollinearity. Right: after grouping into nine interpretable families, all grouped VIFs fall below 5. The grouping reduces the instability enough for coherent family-level interpretation.

The result is reassuring, but the right comparison is in levels rather than percentages. Because train-era preferential-attachment precision is very close to zero, percentage lifts are mechanically unstable. In absolute $P@100$ terms, the reranker still generalizes cleanly forward. At $h = 5$, its gap over preferential attachment rises from +0.045 in the 1990–2005 cells to +0.175 in the held-out 2010–2015 era. At $h = 10$, the gap rises from +0.113 to +0.285. The likely explanation is that the literature graph is thicker in later years, giving the reranker more structure to exploit. But the central conclusion does not depend on that interpretation: the learned graph features do not collapse out of sample when applied to a later era.

### H.9 Regime and horizon checks

These appendix figures answer two interpretation questions that are useful but not central to the main comparison. Figure 31 asks where the transparent graph score gains the most over preferential attachment once the benchmark links are split by local density and by endpoint FWCI. Figure 32 asks how the transparent benchmark changes when the horizon is shortened or extended beyond the paper's main 5, 10, and 15-year design.

## I Credibility Audit Summaries

The main score does not yet fully weight evidence quality, but the benchmark object is not blind to it either. This appendix exists to answer a narrow concern: is the graph mostly noisy co-mention, or does the benchmark sit on a substantively credible directed-claim layer? The extraction layer already records stability, causal presentation, evidence type, and related claim metadata. Tables 18, 19, and 20 should therefore be read as a quality audit of the empirical object rather than as a replacement ranking model. They show that directed causal rows are a smaller but relatively high-stability slice of the graph, that

design-heavy method families remain well represented inside that slice, and that the current benchmark is not built from unstructured co-occurrence counts.

**Table 18:** Credibility audit by edge kind

| Edge kind | Rows | Papers | Mean stability | Explicit-causal share |
|---|---|---|---|---|
| Directed causal | 89,737 | 23,213 | 0.930 | 69.0% |
| Undirected contextual | 1,181,277 | 221,192 | 0.868 | 45.3% |

*Notes.* Each row summarizes one edge layer in the extracted graph. "Rows" is the number of extracted relations, "Papers" is the number of papers containing at least one such relation, "Mean stability" is the extraction-model stability score averaged across rows, and "Explicit-causal share" is the share of rows whose language is coded as explicitly causal. The point is descriptive: the directed layer is smaller, but it is also the more stable and more explicitly causal slice of the graph.

**Table 19:** Directed-causal credibility audit by evidence type

| Evidence type | Rows | Mean stability | Explicit-causal share |
|---|---|---|---|
| Panel FE / TWFE | 44,106 | 0.938 | 68.8% |
| Difference-in-differences | 16,658 | 0.933 | 75.1% |
| Experiment | 15,616 | 0.900 | 56.7% |
| Event study | 6,247 | 0.940 | 73.5% |
| Instrumental variables | 5,601 | 0.933 | 78.3% |
| Regression discontinuity | 1,509 | 0.922 | 80.3% |

*Notes.* This table restricts attention to directed-causal rows and then splits them by the main evidence method recorded in the extraction metadata. "Rows" is the number of extracted directed-causal relations in that method family. "Mean stability" and "Explicit-causal share" are row-level averages within the method family. The table shows that design-heavy empirical methods remain well represented inside the directed layer.

**Table 20:** Stability bands by edge kind

| Edge kind | High stability | Mid stability | Low stability |
|---|---|---|---|
| Directed causal | 91.8% | 5.6% | 2.6% |
| Undirected contextual | 85.4% | 5.1% | 9.5% |

*Notes.* Each row shows how extraction rows are distributed across the paper's three stability bands. The percentages sum to 100 within edge kind. The comparison is included to show that low-stability rows are relatively uncommon overall and especially uncommon in the directed-causal layer.

# J Supplementary Usefulness Validation

These checks ask a different question from the historical benchmark. The historical benchmark asks whether a graph-grounded question later becomes part of the literature. The usefulness checks ask whether the rendered question looks useful and intelligible to a current reader. They speak to presenta-

tion quality and semantic coherence, not to historical forecasting. They remain in the appendix because they assess the surfaced question, not because they replace the benchmark.

**Pending today.** The paired-family LLM usefulness screen at the current baseline path setting is now complete off-paper. Once the parked path-length axis has identified the best longer-path setting, this appendix section should be refreshed to report the paired-family comparison and then one selective length comparison between ‘max_path_len = 2‘ and the best longer-path contender, rather than all four path-length settings.

The human exercise uses a small blinded pack built from the path-to-direct frontier. It compares 12 graph-selected items with 12 preferential-attachment-selected items drawn from the same candidate universe, with repeated sources and targets capped within arm so the pack does not collapse into one crowded endpoint neighborhood. Each item is shown as a compact graph-grounded question object plus a short construction note, and raters score readability, interpretability, usefulness, and artifact risk. Readability is intentionally narrow: it measures whether the labels are clean and easy to parse, not whether the underlying idea is substantively better. On this 24-item pack, the overall mean score is identical across arms at 3.22. Graph-selected items score somewhat higher on interpretability (3.17 versus 2.92), slightly higher on usefulness (3.00 versus 2.92), and slightly lower on artifact risk (1.58 versus 1.67), while preferential-attachment items score higher on readability (3.83 versus 3.50). The right reading is cautious external validation, not a broad human-rated advantage.

The appendix LLM usefulness sweep applies a related but more compressed question object on a much larger grid. The model sees a simplified graph-grounded question object plus a short construction note. That makes the exercise useful as a coarse screening check for readability and artifact risk, but not as a substitute for the historical benchmark or for the richer current-frontier objects used in the human exercise. The model is explicitly instructed not to judge novelty at the cutoff, likely future success, or topic importance. In practice, this comparison is informative mainly at the level of broad screening differences across arms rather than as a fine ranking of individual questions. For that reason, it is reported only as supplementary appendix evidence.

# K Additional Analysis

## K.1 Bundle uptake

The main paper treats one predicted edge as the unit of analysis. This section asks whether the downstream unit is often larger. When a later paper realizes one historically predictable edge, does it usually realize that edge alone, or does it take up a small bundle of nearby graph-supported ideas in the same paper?

We build a historical edge-to-paper uptake spine that links each realized in-pool prediction to the later paper or papers that realize it. For *path-to-direct*, the realizing paper is the paper that first writes the direct causal edge. For *direct-to-path*, the realizing paper is the paper that first supplies the path event. We then

| horizon | n_papers | mean_predicted_edge_count | share_multi_edge | share_mixed_family | share_shared_endpoint | share_shared_mediator | median_min_graph_distance |
|---|---|---|---|---|---|---|---|
| 5 | 1779 | 1.054525 | 0.051152 | 0.001124 | 0.039348 | 0.023609 | 0.000000 |
| 10 | 2277 | 1.055336 | 0.052701 | 0.001757 | 0.040843 | 0.023276 | 0.000000 |
| 15 | 2638 | 1.055345 | 0.053071 | 0.001516 | 0.041698 | 0.022745 | 0.000000 |

**Table 21:** Bundle uptake summary by horizon.

deduplicate to the first realizing paper for each prediction-horizon pair and aggregate to the paper level. For each later paper we count how many historically predicted edges it realizes, whether those realized edges come from one family or both, and whether the realized edges are close in the earlier graph. Distance is measured on the historical support graph available at the relevant cutoff.

Table 21 shows a clear pattern. Most later papers realize exactly one historically predicted edge. Across horizons, only about five percent of realizing papers realize more than one predicted edge in the same paper. Mixed-family bundles are rarer still: only about 0.1 to 0.2 percent of all realizing papers, or about 2 to 3 percent of the multi-edge subset. The downstream object in this setting therefore still looks like a single paper-shaped question, not a broad mixed package.

When multi-edge uptake does happen, however, it is local rather than diffuse. Among the multi-edge papers, about 77 to 79 percent share at least one endpoint across realized predicted edges, about 43 to 46 percent share a mediator, and the median minimum graph distance is zero at every horizon. Figure 33 shows both the dominance of single-edge uptake and the fact that the small set of multi-edge bundles is overwhelmingly local and mostly within-family.

The main implication is interpretive. The budget result in Appendix B.3 does not seem to be driven by large numbers of later papers absorbing broad mixed bundles of predicted ideas. Instead, the wider-budget advantage of *direct-to-path* appears to come mainly from broader coverage across many separate later papers. When a later paper does realize more than one predicted edge, those edges tend to be tightly local in graph space.

## K.2 Adopter profiles

This section asks which kinds of papers and teams independently move toward the same graph-supported questions. The exercise is descriptive rather than causal. The claim is not that these adopters use Frontier Graph. The question is whether different question families tend to be taken up by different kinds of scientific actors.

We start from the first-realizing-paper version of the uptake spine and merge it with paper metadata, authorship records, and a targeted OpenAlex enrichment pass. Local corpus data provides paper year, subfield, venue, funding fields, team size, and local author history. Targeted OpenAlex API pulls supply focal-paper affiliation country and institution type, as well as author-level counts-by-year that we use to recover global career age and prior output up to the focal paper year. Coverage is high for the author-history variables (missing share about 0.2 percent) and good enough for the focal affiliation variables (missing share about 11 to 12 percent).

| horizon | candidate_family_mode | n_papers | mean_team_size | share_any_funder | share_cross_country | share_incumbent_local | share_bridge_author_local | mean_career_age_local | mean_career_age_global | mean_prior_works_global | share_mixed_family |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 5 | direct_to_path | 1638 | 2.230000 | 0.034000 | 0.182000 | 0.539000 | 0.170000 | 6.163000 | 21.868000 | 55.001000 | 0.001000 |
| 5 | path_to_direct | 143 | 2.804000 | 0.140000 | 0.294000 | 0.530000 | 0.146000 | 5.758000 | 15.862000 | 55.057000 | 0.014000 |
| 10 | direct_to_path | 2038 | 2.388000 | 0.054000 | 0.204000 | 0.535000 | 0.167000 | 6.299000 | 22.092000 | 58.243000 | 0.002000 |
| 10 | path_to_direct | 243 | 2.856000 | 0.185000 | 0.300000 | 0.544000 | 0.136000 | 5.914000 | 18.301000 | 59.829000 | 0.016000 |
| 15 | direct_to_path | 2306 | 2.477000 | 0.077000 | 0.213000 | 0.533000 | 0.165000 | 6.300000 | 22.028000 | 61.400000 | 0.002000 |
| 15 | path_to_direct | 336 | 2.789000 | 0.182000 | 0.280000 | 0.539000 | 0.127000 | 5.766000 | 18.269000 | 59.275000 | 0.012000 |

**Table 22:** Adopter-profile overview by horizon and family.

| Horizon | Metric | Path-direct | Direct-path | Difference | CI low | CI high | p | N path | N direct |
|---|---|---|---|---|---|---|---|---|---|
| 5 | Team size | 2.804000 | 2.230000 | 0.574000 | 0.137000 | 1.012000 | p=0.010 | 143 | 1636 |
| 5 | Any recorded funder | 0.140000 | 0.034000 | 0.106000 | 0.048000 | 0.163000 | p<0.001 | 143 | 1638 |
| 5 | Cross-country team | 0.294000 | 0.182000 | 0.112000 | 0.035000 | 0.189000 | p=0.004 | 143 | 1636 |
| 5 | Share incumbent authors | 0.530000 | 0.539000 | -0.008000 | -0.072000 | 0.055000 | p=0.794 | 143 | 1628 |
| 5 | Share bridge authors | 0.146000 | 0.170000 | -0.025000 | -0.080000 | 0.031000 | p=0.389 | 121 | 1253 |
| 5 | Mean global career age | 15.862000 | 21.868000 | -6.006000 | -7.886000 | -4.126000 | p<0.001 | 143 | 1628 |
| 5 | Mean global prior works | 55.057000 | 55.001000 | 0.056000 | -8.808000 | 8.920000 | p=0.990 | 143 | 1628 |
| 10 | Team size | 2.856000 | 2.388000 | 0.468000 | 0.161000 | 0.775000 | p=0.003 | 243 | 2035 |
| 10 | Any recorded funder | 0.185000 | 0.054000 | 0.131000 | 0.081000 | 0.181000 | p<0.001 | 243 | 2038 |
| 10 | Cross-country team | 0.300000 | 0.204000 | 0.096000 | 0.036000 | 0.157000 | p=0.002 | 243 | 2035 |
| 10 | Share incumbent authors | 0.544000 | 0.535000 | 0.008000 | -0.041000 | 0.058000 | p=0.738 | 243 | 2026 |
| 10 | Share bridge authors | 0.136000 | 0.167000 | -0.031000 | -0.074000 | 0.012000 | p=0.155 | 206 | 1580 |
| 10 | Mean global career age | 18.301000 | 22.092000 | -3.791000 | -5.550000 | -2.032000 | p<0.001 | 243 | 2026 |
| 10 | Mean global prior works | 59.829000 | 58.243000 | 1.585000 | -6.603000 | 9.774000 | p=0.704 | 243 | 2026 |
| 15 | Team size | 2.789000 | 2.477000 | 0.312000 | 0.073000 | 0.551000 | p=0.010 | 336 | 2302 |
| 15 | Any recorded funder | 0.182000 | 0.077000 | 0.104000 | 0.062000 | 0.147000 | p<0.001 | 336 | 2306 |
| 15 | Cross-country team | 0.280000 | 0.213000 | 0.066000 | 0.016000 | 0.117000 | p=0.010 | 336 | 2302 |
| 15 | Share incumbent authors | 0.539000 | 0.533000 | 0.006000 | -0.037000 | 0.049000 | p=0.799 | 335 | 2292 |
| 15 | Share bridge authors | 0.127000 | 0.165000 | -0.038000 | -0.075000 | -0.001000 | p=0.044 | 281 | 1801 |
| 15 | Mean global career age | 18.269000 | 22.028000 | -3.759000 | -5.309000 | -2.208000 | p<0.001 | 335 | 2292 |
| 15 | Mean global prior works | 59.275000 | 61.400000 | -2.124000 | -9.371000 | 5.123000 | p=0.566 | 335 | 2292 |

**Table 23:** Descriptive difference tests for the adopter-profile comparison.

Table 22 shows the main pattern. Relative to *direct-to-path*, papers that realize *path-to-direct* predictions tend to have larger teams, higher funding incidence, and more cross-country collaboration. At $h = 10$, for example, *path-to-direct* adopters average 2.86 authors versus 2.39 for *direct-to-path*; 18.5 percent versus 5.4 percent have any recorded funder; and 30.0 percent versus 20.4 percent are cross-country teams. Table 23 shows that these differences are statistically precise in simple large-sample comparisons. Incumbent-share and bridge-share differences are not. *Path-to-direct* adopters are also somewhat younger in global career age.

The team-composition split in Figure 34 sharpens that difference. *Path-to-direct* uptake is much less likely to be solo-authored and more likely to come from teams of two to six authors. The outlet mix differs too, but the difference sits within economics rather than between economics and non-economics. *Path-to-direct* uptake tilts more toward energy, environment, development, and applied-health outlets such as *Sustainability*, *Energy Policy*, *Energy Economics*, *World Development*, and the *Journal of Development Economics*. *Direct-to-path* uptake is relatively more present in canonical economics, finance, and health-policy outlets such as the *American Economic Review*, *Journal of Banking & Finance*, *Journal of Economic Dynamics and Control*, *The Journal of Finance*, *Health Affairs*, and *Health Economics*. The top-five general-interest economics journals remain a small share in both families, so the difference is not driven by a handful of flagship outlets. The cleaner venue-tier split confirms that both families are overwhelmingly in economics-facing

journals, with only a small adjacent share. The sharper separation is in team structure and resource intensity, with a secondary tilt in venue content and topic composition. The paper-type split points in the same direction: *path-to-direct* is relatively more concentrated in energy and environment, while *direct-to-path* is relatively more concentrated in finance, health-policy, and theory-or-methods papers.

These descriptives suggest a plausible division of labor across the two question families. *Path-to-direct* ideas appear to be taken up by somewhat more coordinated and better-resourced teams, with more cross-border collaboration and slightly lower mean global career age. *Direct-to-path* uptake is broader and more diffuse: it appears in more papers overall, but more often in smaller or solo teams and less often in funded projects. If that pattern survives fuller work, it would sharpen the interpretation from the budget analysis. The two families would differ not only in what they surface, but also in which scientific actors tend to move toward them on their own.

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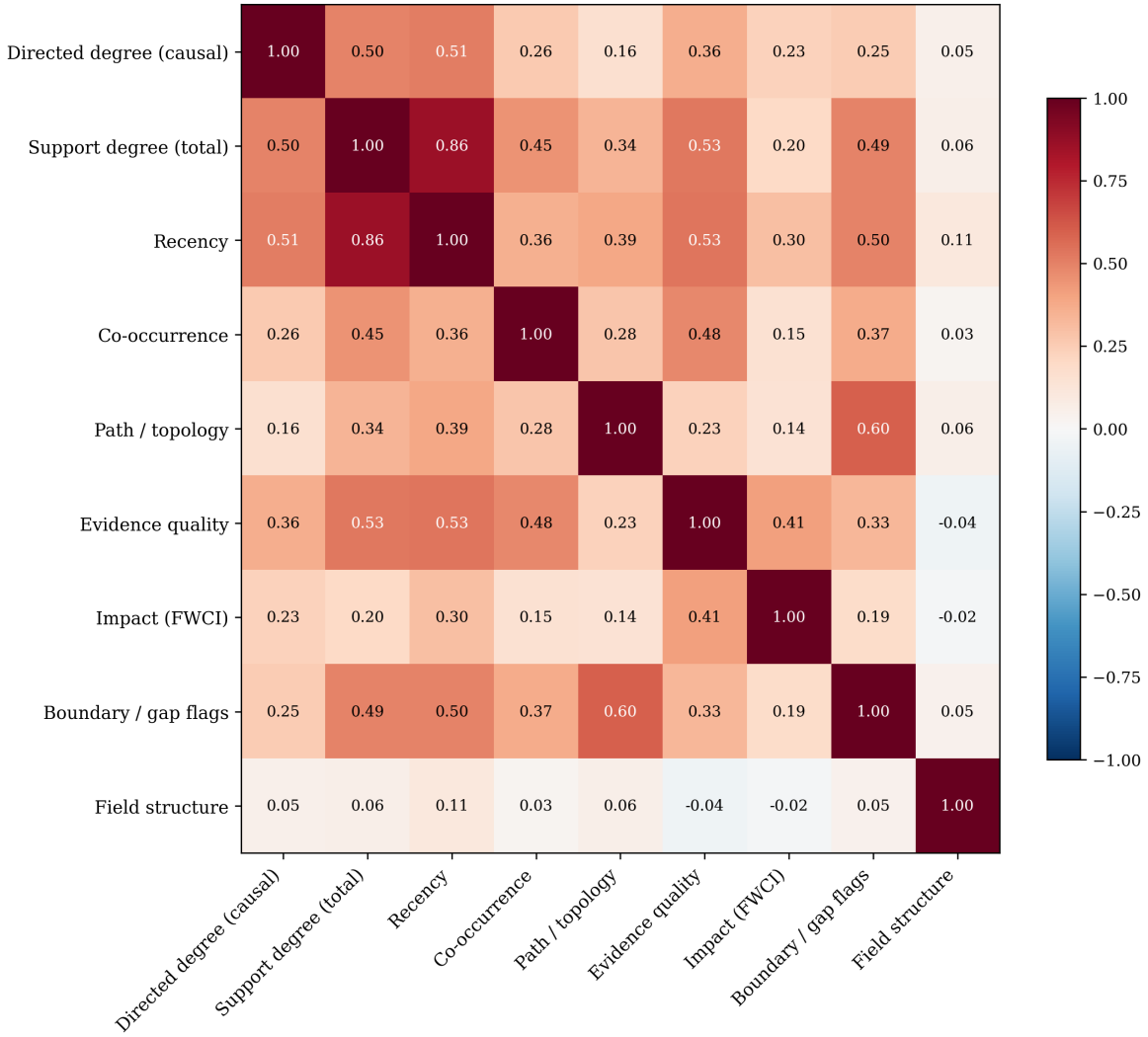
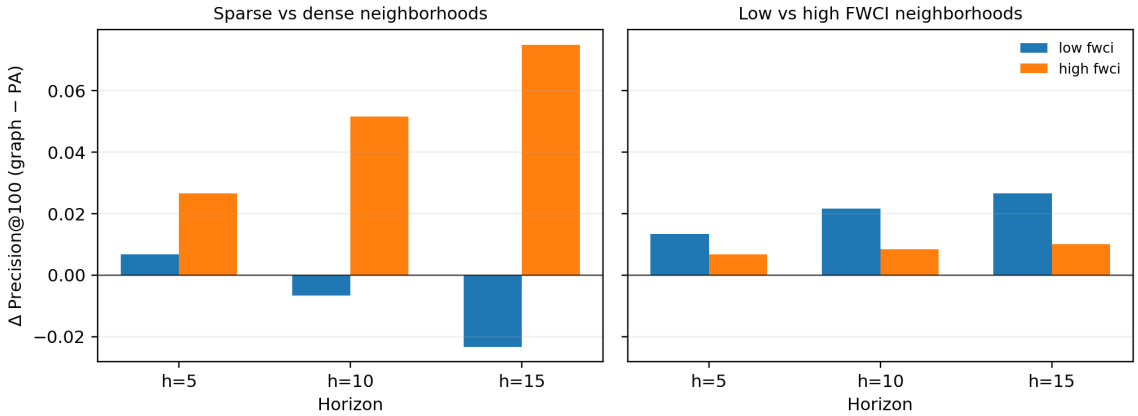
**Figure 30:** Correlations across grouped feature families



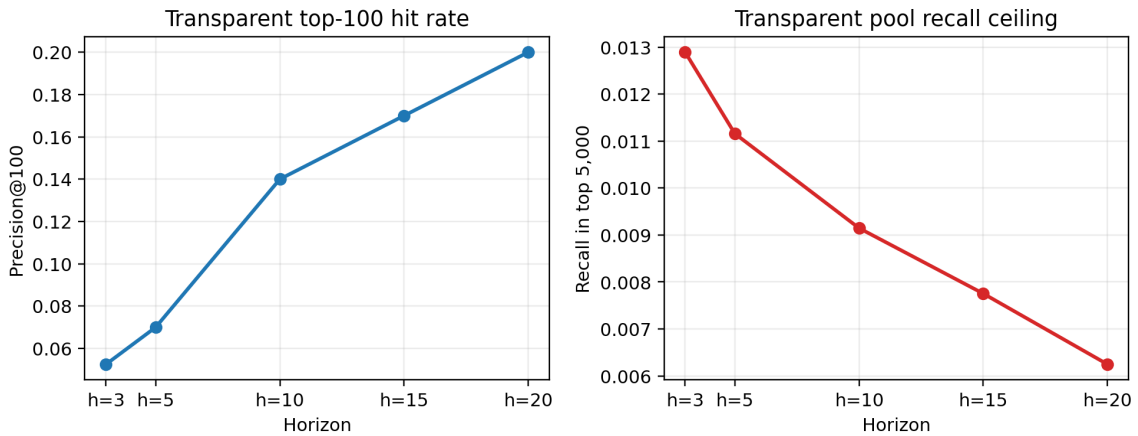

*Notes.* Pairwise correlations between the nine group-level PC1 scores. The strongest remaining correlation is between support degree and recency (about 0.86), so the grouped features are not orthogonal. But the grouped VIFs remain below 5, which is enough for substantially cleaner interpretation than at the raw-feature level.

**Figure 31:** The transparent score helps more in dense neighborhoods




*Notes.* Each bar reports the transparent score's mean Precision@100 advantage over preferential attachment on the main 1990–2015 benchmark panel, so positive values mean more later-realized links per 100 surfaced candidates. The left panel splits candidate pairs by whether they sit in denser or sparser local co-occurrence neighborhoods within the same benchmark cell. The right panel splits them by pair-level field-weighted citation impact (FWCI). The figure is included to show where the graph helps, not just whether it helps. Its edge is largest in denser local neighborhoods, and it is relatively larger in lower-FWCI slices than in the highest-FWCI ones.

**Figure 32:** Longer horizons raise hit rates, but the graph remains a screening layer




*Notes.* This figure uses the same fixed 5,000-candidate pool at each horizon. The left panel reports later-realized links per 100 surfaced candidates. The right panel reports Recall@100, the share of all later-realized links captured in the top 100 candidates, together with the much higher recall available if one inspected the full candidate pool. As the horizon lengthens, top-100 hit rates rise because more future links become eligible realizations. But Recall@100 remains small relative to the full-pool ceiling, which is why the paper treats the transparent score as a screening layer and studies reranking and reading budgets separately.

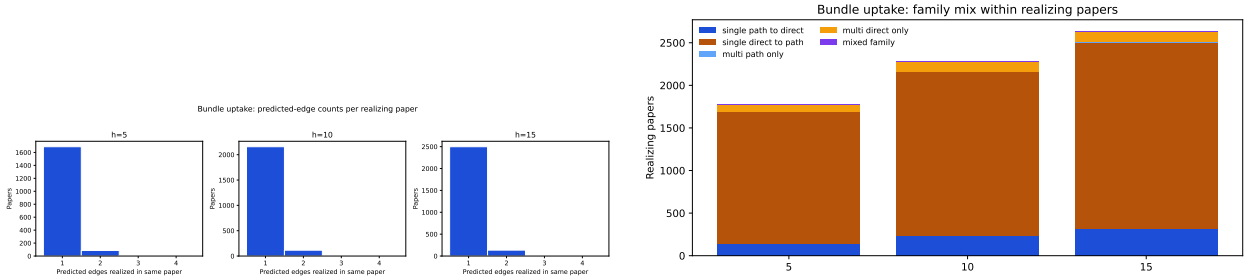



**Figure 33:** Bundle uptake descriptives. Left: the number of historically predicted edges realized in the same later paper is almost always one. Right: the small multi-edge subset is dominated by within-family bundles, especially *direct-to-path*.

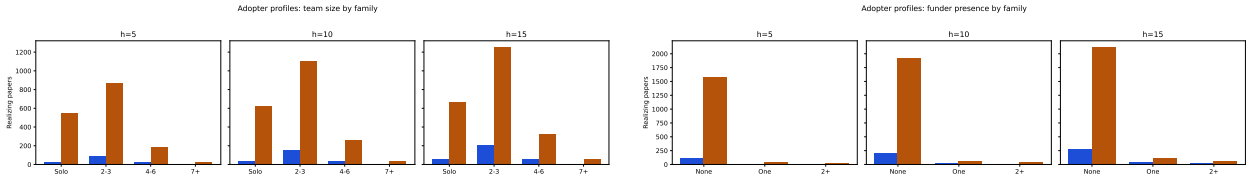



**Figure 34:** Adopter-profile descriptives. Left: *path-to-direct* uptake is less often solo-authored and more concentrated in two-to-six-author teams. Right: *path-to-direct* uptake is more likely to appear in papers with recorded funding.